

MANTHRA-X: PIONEERING PRECISION, THE FUTURE OF AUTONOMOUS MOBILITY

Project ID; 24_25J_213



MEMBERS



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INTRODUCTION



BACKGROUND



Currently Available

Waymo's Perception System

- Waymo uses a combination of LiDAR, cameras, and radar with advanced CNNs and RNNs for object detection and scene understanding.
- Limitations:
 - Data Requirements: High dependency on large volumes of labeled data for training.
 - Computational Resources: Requires substantial computational power for real-time processing.
 - Generalization: May struggle with scenes not well-represented in training data and not suitable for under-developed infrastructure.



Currently Available

Tesla's Autopilot

- Tesla's system uses camera-based vision with deep learning models to interpret the environment and make driving decisions.
- Limitations:
 - Weather Conditions: Performance can degrade in adverse weather conditions.
 - Edge Cases: Difficulty handling rare or unusual driving scenarios.



Currently Available

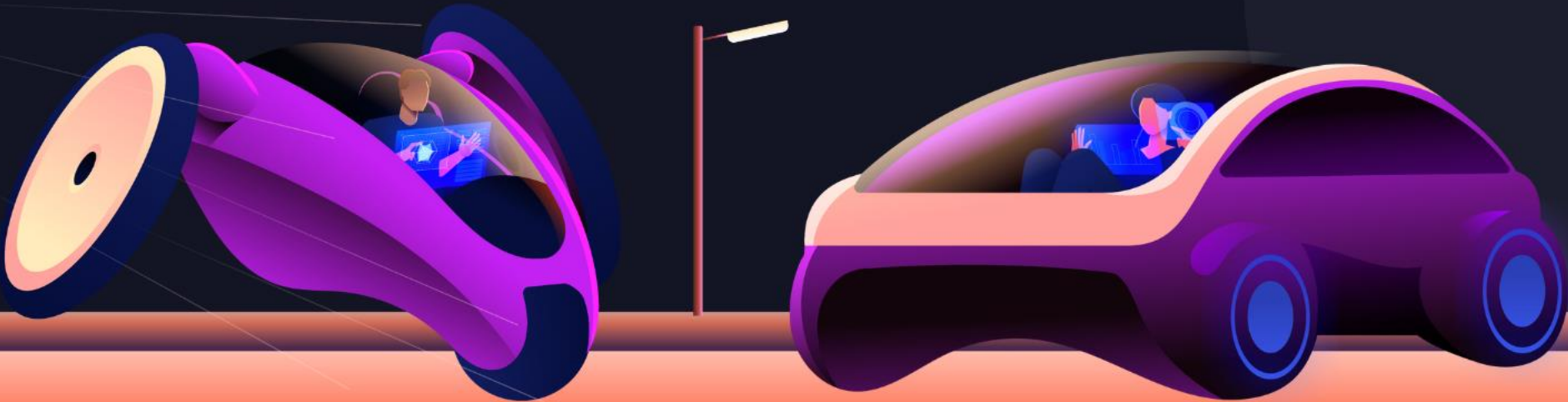
Nuro's Autonomous Delivery Vehicles

- Employs reinforcement learning and planning algorithms for navigation and collision avoidance. Optimized for specific tasks like delivery routes with a focus on efficiency and safety.
- Limitations:
 - Limited Context: Designed primarily for delivery routes, not general driving environments.
 - Predictive Accuracy: May struggle with predicting and adapting to human drivers' unpredictable behaviors.



OBJECTIVE

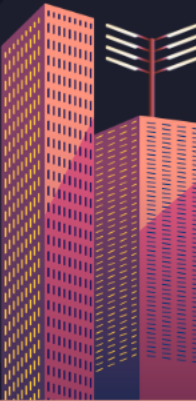
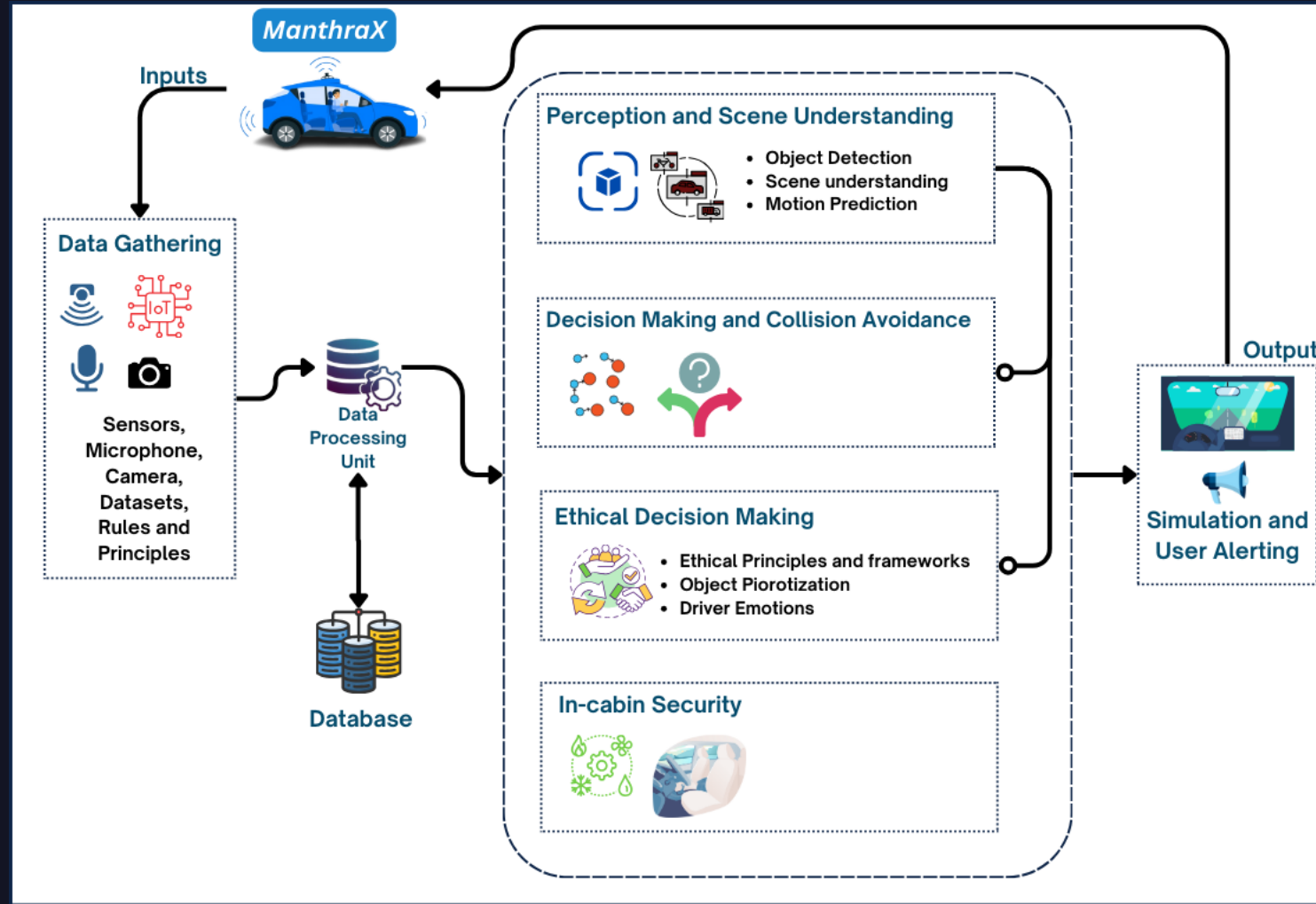
Manthra-X is enhancing perception and scene understanding, decision-making, and in-cabin security. This involves integrating advanced machine learning models and simulations to improve vehicle safety, ethical decision-making, and passenger comfort in autonomous systems.



PROPOSED SOLUTION



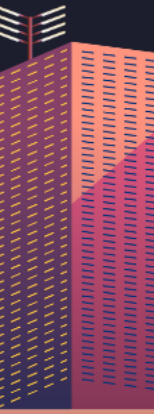
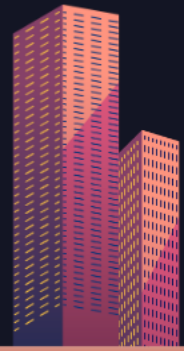
Overall System Diagram





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BSc (Hons) Degree in Information Technology (specialization in Data Science)
DATA SCIENCE

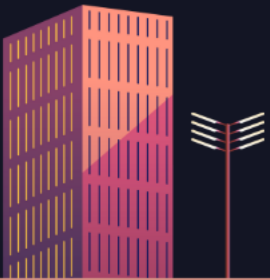
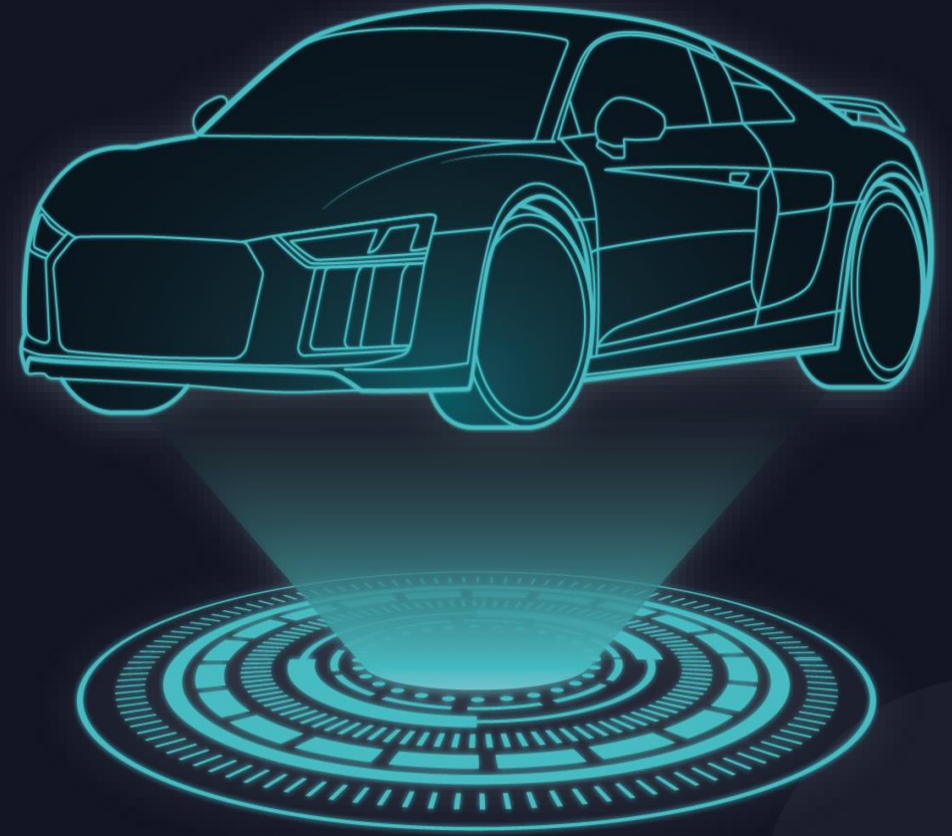


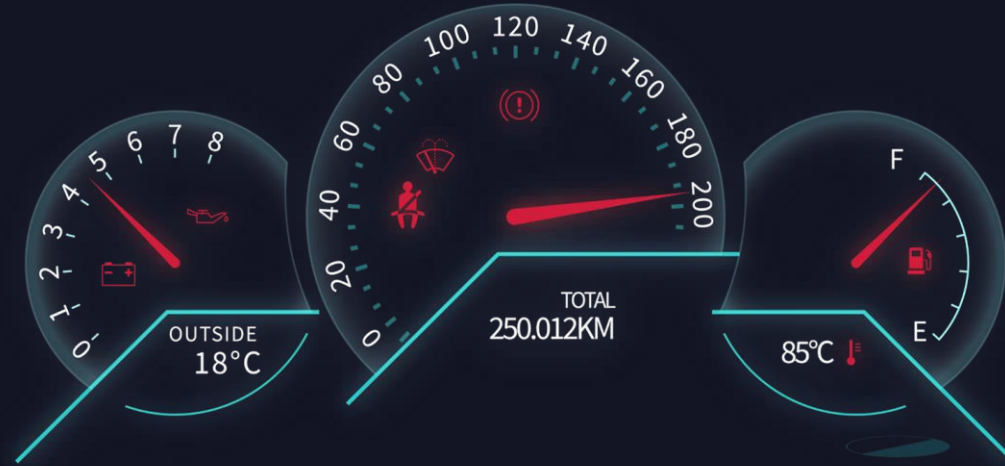
Component 01

PERCEPTION AND SCENE UNDERSTANDING



Enhancing Perception and Motion Prediction in Autonomous Vehicles Through Hybrid Modeling and Self-Supervised Learning.





BACKGROUND

Why Enhancing perception and Scene Understanding?

Improved
Safety

Data Efficiency

Expanded
Capabilities

Enhanced
Driving
Experience

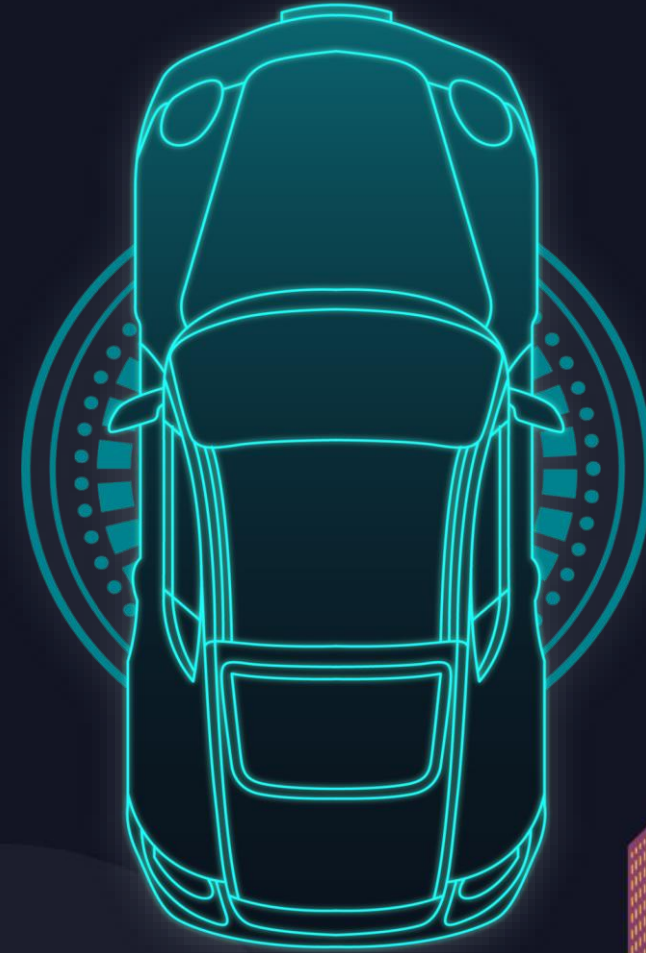


EXISTING AUTONOMOUS VEHICLES LACK,

Complex
Scenario
Handling

Sensor
Improvements

Minimized
Data
Dependency



RESEARCH GAP

Research 01
**"TransVOD:
Transformer-Based
Visual Object Detection
for Self-Driving Cars"**

Research 02
**"TrackFormer: Multi-
Object Tracking with
Transformers:"**

Research 03
**"Transformers in
Autonomous Driving: A
Survey"**

Research 04
**"Dynamic Graph Neural
Networks for Modeling
Object Interactions"**

Research 05
**"Self-Supervised Learning
for Object Detection and
Tracking"**

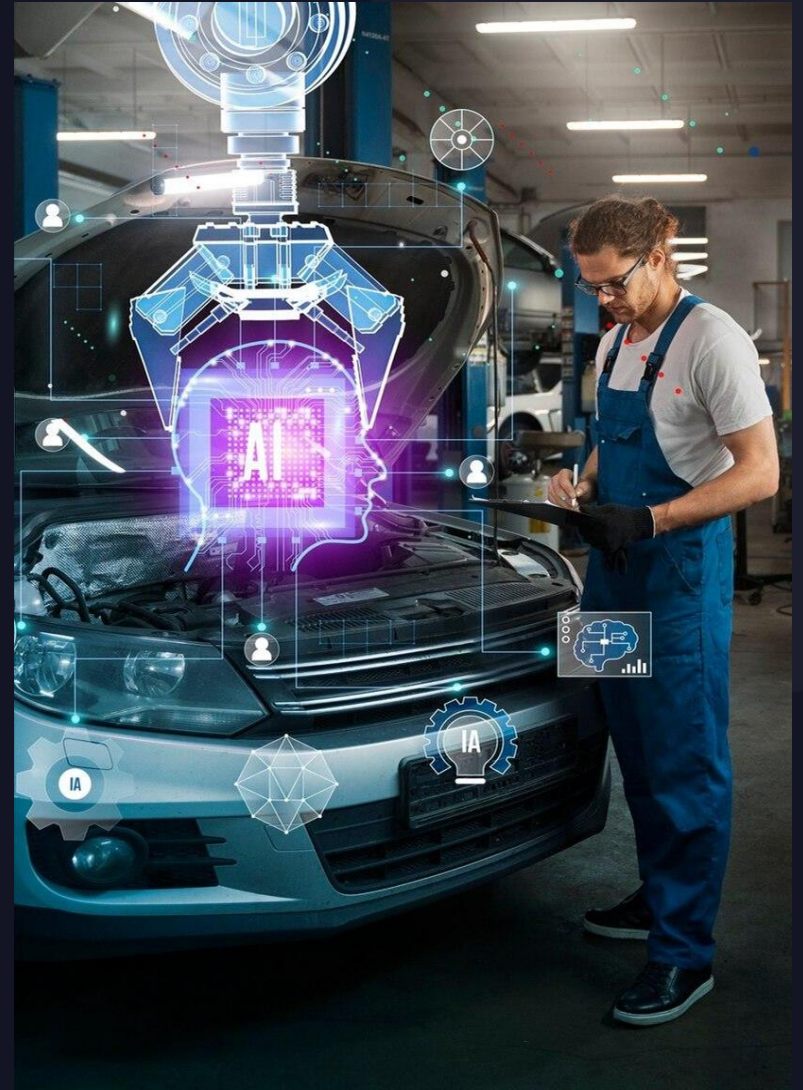
RESEARCH GAP

Features	Research 1	Research 2	Research 3	Research 4	Research 5	MANTHRA-X
Modeling of Spatial relationship between objects.	No	No	No	No	No	
Address both spatial relationships and temporal dependencies in object detection.	No	No	No	No	No	
Improved accuracy in occlusions and object deformations.	No	Yes	No	No	No	
Enhanced model's learning capability from unlabeled data.	No	No	No	No	Yes	
Combines different sensor modelities.	Yes	Yes	No	No	No	



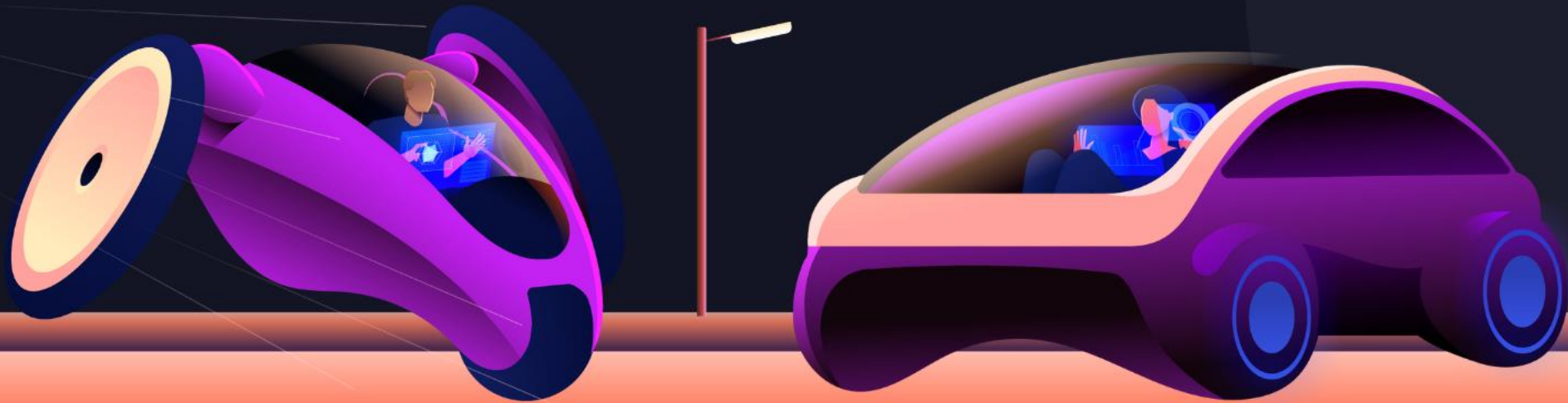
RESEARCH PROBLEM?

- How can graph neural networks be effectively integrated with transformer architectures for multi-modal sensor fusion in autonomous vehicles?
- What self-supervised tasks are most effective for improving object detection and motion prediction in challenging driving conditions?
- Can self-supervised pre-training enhance the performance of hybrid models for autonomous vehicle perception?



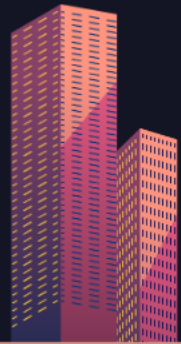
SPECIFIC OBJECTIVE

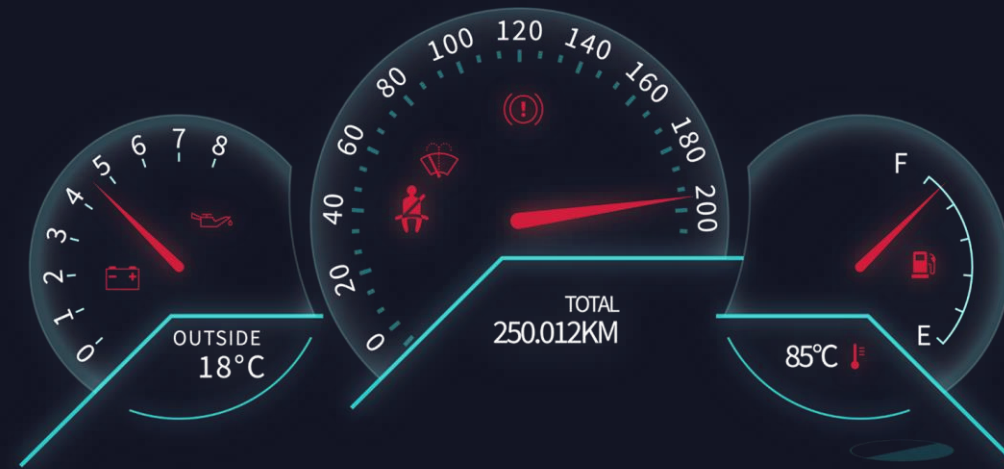
- ❖ To develop a robust and efficient perception system for autonomous vehicles capable of accurately detecting, tracking, and predicting the behavior of objects in complex urban environments by integrating hybrid model architectures and self-supervised learning techniques.



SUB-OBJECTIVES

- ❖ **Develop a hybrid model architecture:** Design and implement a hybrid model combining transformer and graph neural network components to effectively fuse multi-modal sensor data for enhanced object perception and tracking.
- ❖ **Improve object detection and tracking:** Enhance object detection and tracking performance in challenging conditions, such as occlusions, low visibility, and dynamic environments.
- ❖ **Advance motion prediction:** Develop accurate motion prediction models capable of predicting the future trajectories of objects, including pedestrians, vehicles, and cyclists.
- ❖ **Optimize for real-time performance:** Design and implement efficient algorithms and data structures to enable real-time operation of the perception system.
- ❖ **Leverage self-supervised learning:** Explore effective self-supervised learning techniques to improve model generalization and reduce reliance on labeled data.

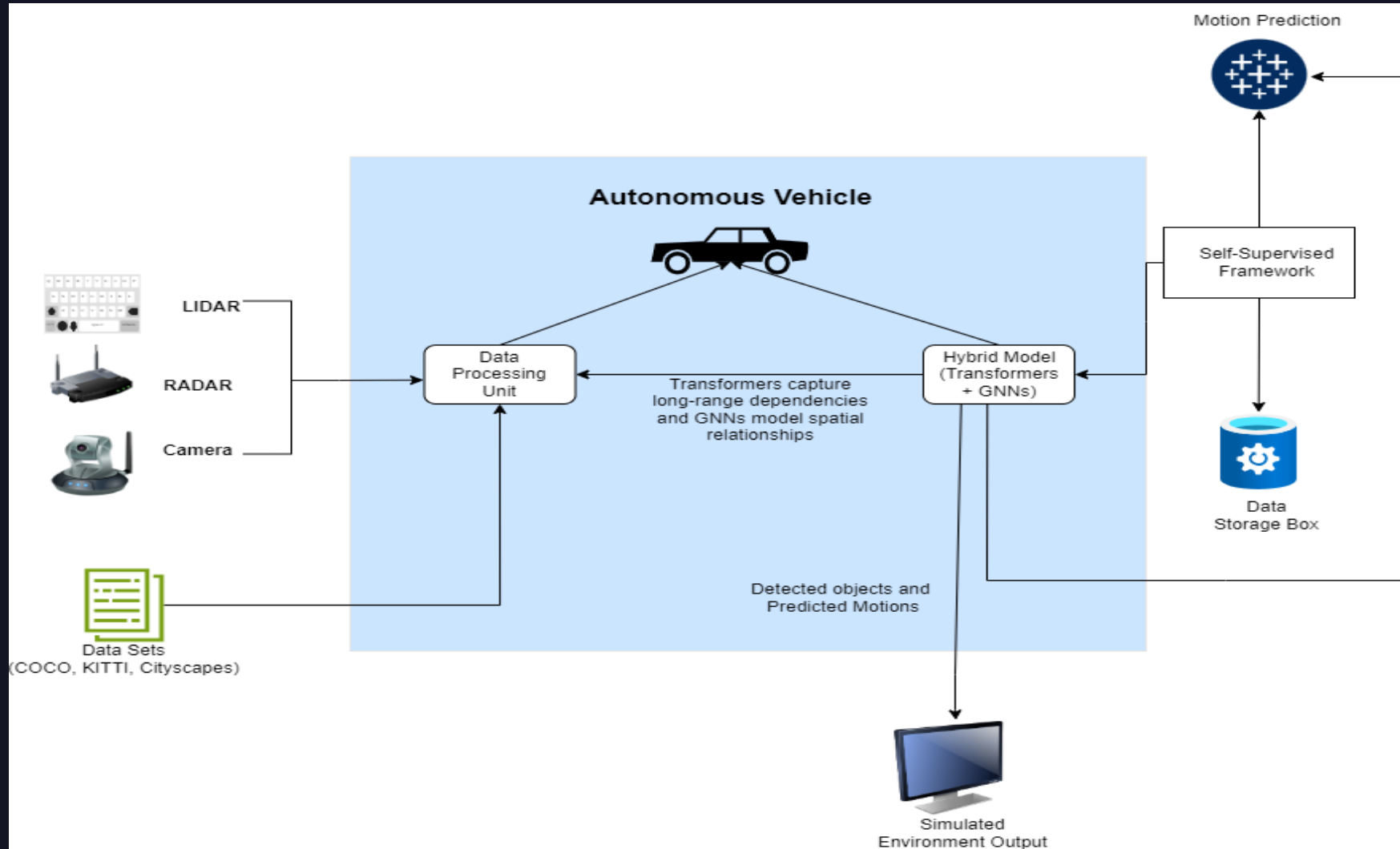




METHODOLOGY



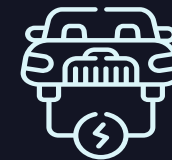
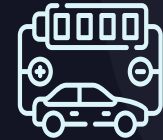
SYSTEM DIAGRAM



REQUIREMENT SPECIFICATION

System Requirements

- ❖ **High-performance computing (HPC) cluster or workstations:** For model training, simulation, and real-time processing.
- ❖ **LiDAR sensors:** For accurate distance and object detection.
- ❖ **Radar sensors:** For object detection and velocity estimation in adverse weather conditions.
- ❖ **High-resolution cameras:** For capturing detailed visual information.
- ❖ **IMU (Inertial Measurement Unit):** For vehicle motion sensing and stabilization.



REQUIREMENT SPECIFICATION

Software Requirements

Development Tools:

IDEs: VSCode, PyCharm, Jupyter Notebook
Version Control: Git(GitHub, GitLab)

Frameworks and Libraries:

AI/ML: Tensorflow, Pytorch, Keras
Libraries: OpenCV, PCL, RADAR Data processing libraries

Simulation Frameworks:
CARLA, AirSim, or Unity

Databases:

NoSQL Databases: MongoDB, Cassandra

Cloud Services:

Microsoft Azure for Computing, storage, and machine Learning

Security: Cryptography Libraries

Web Frameworks: Node.js, Flask, Django

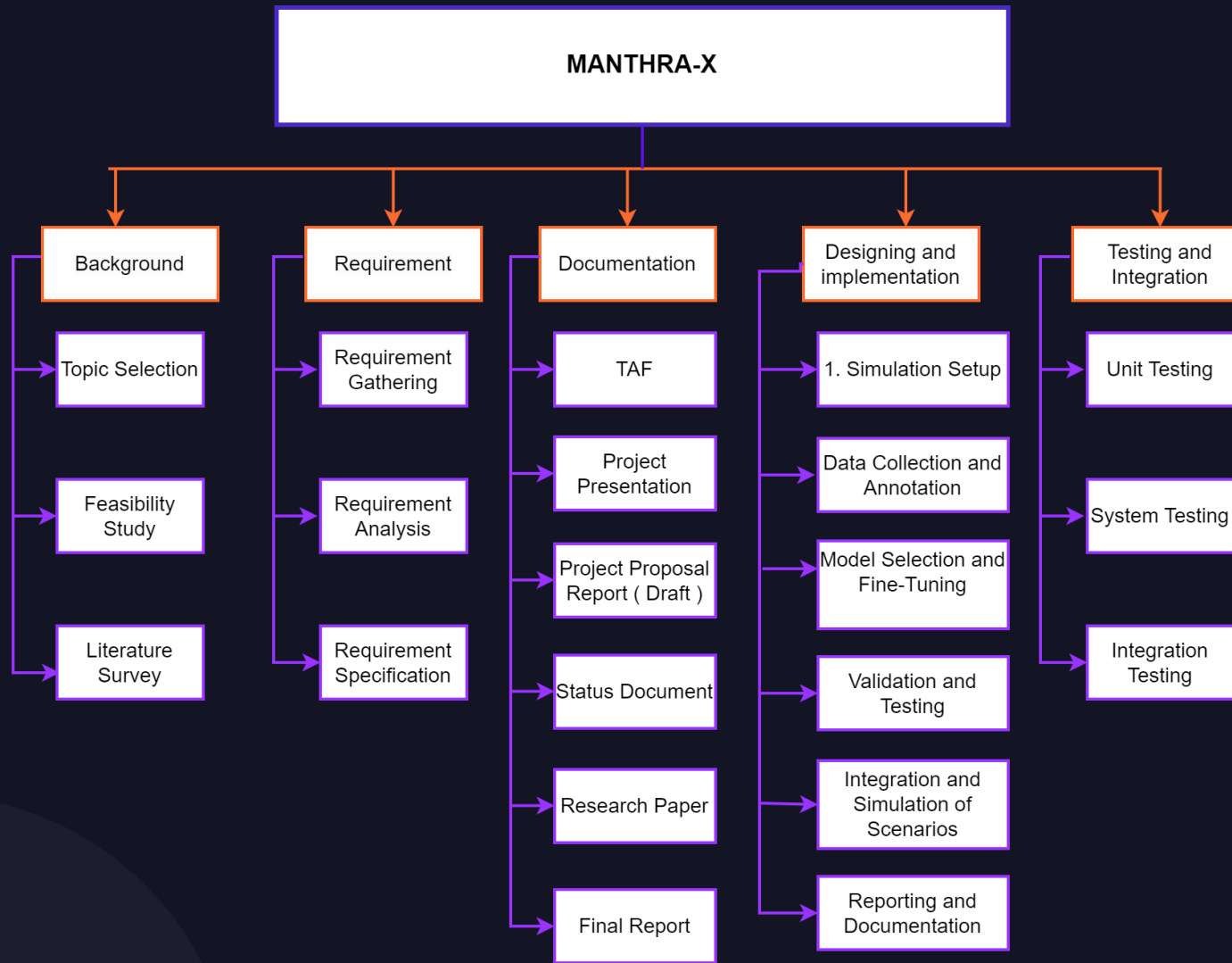


REQUIREMENT SPECIFICATION

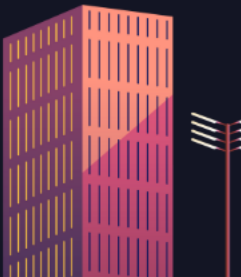
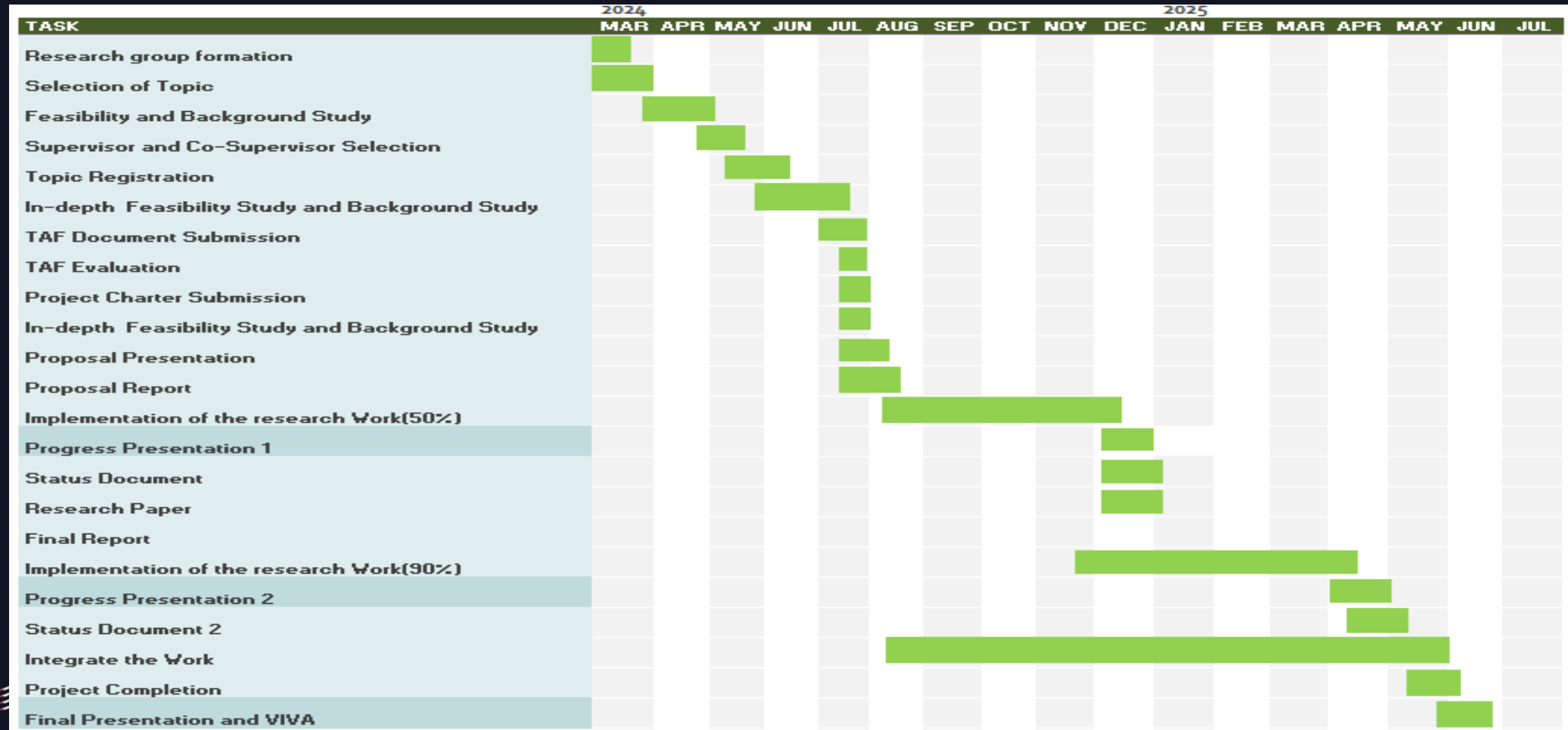
Personal Requirements

- ❖ Skills and Expertise.
- ❖ Time Management.
- ❖ Continuous Learning.
- ❖ Collaboration and Communication.
- ❖ Project Management





GANNT CHART



REFERENCES

[1]Iyoti Madake;Tejas Lokhande;Atharv Mali;Nachiket Mahale;Shripad Bhatlawande, “**TransVOD: Transformer-Based Visual Object Detection for Self-Driving Cars**”, 2024 International Conference on Current Trends in Advanced Computing (ICCTAC)Year: 2024 | Conference Paper | Publisher: IEEE

[2]Tim Meinhardt;Alexander Kirillov;Laura Leal-Taixé;Christoph Feichtenhofer, “**TrackFormer: Multi-Object Tracking with Transformers:**”, 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Year: 2022 | Conference Paper | Publisher: IEEE

[3]"Transformers in Autonomous Driving: A Survey"

[4]"Dynamic Graph Neural Networks for Modeling Object Interactions"

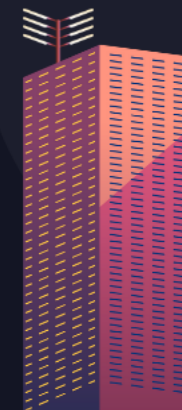
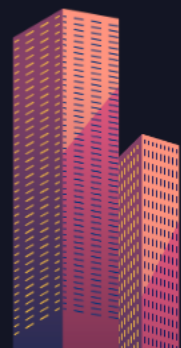
[5]"Self-Supervised Learning for Object Detection and Tracking"





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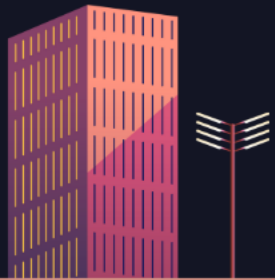
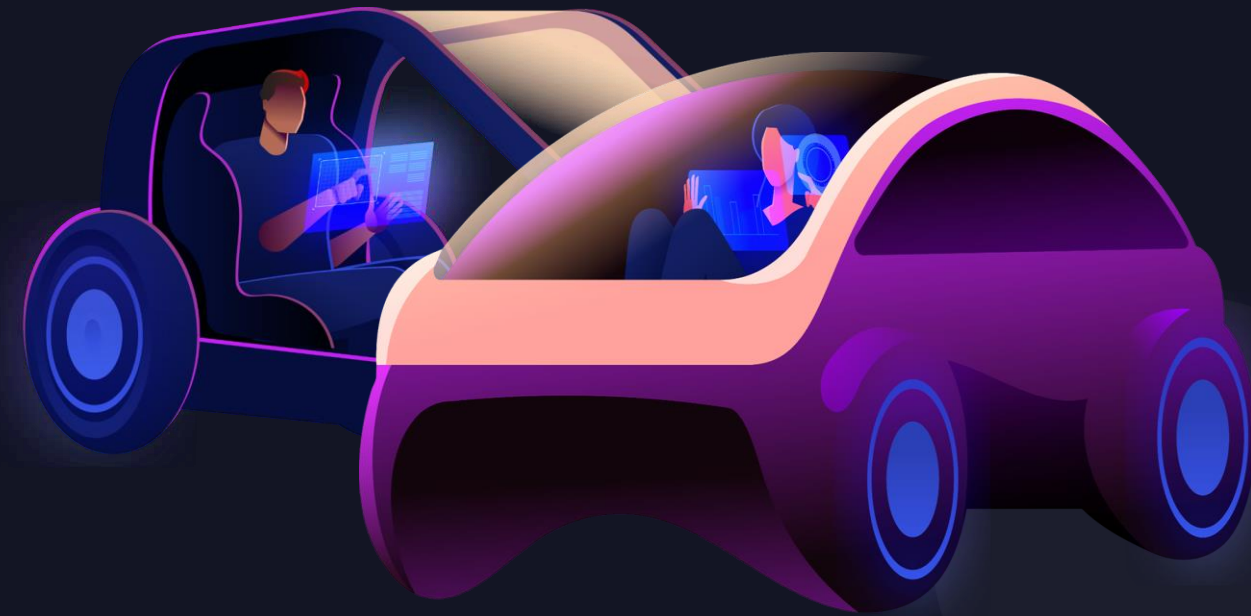


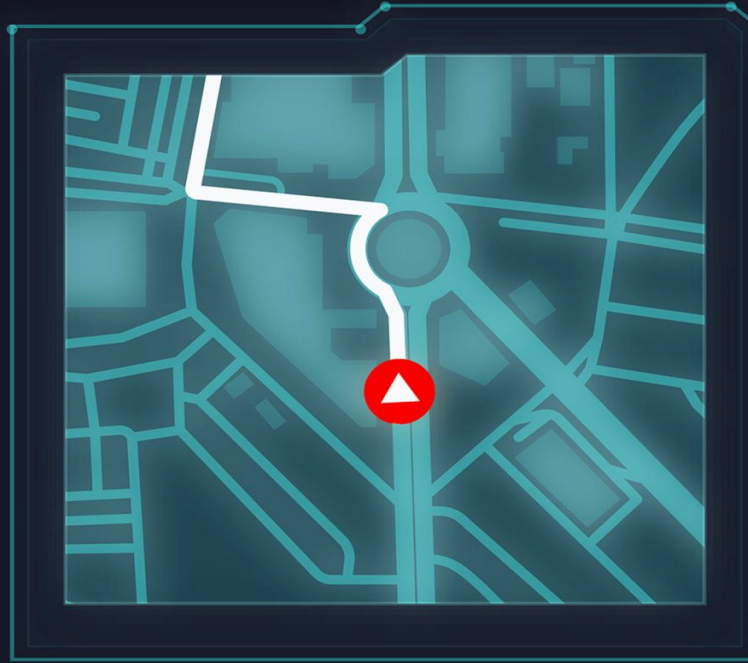
Component 02

DECISION MAKING AND COLLISION AVOIDANCE



Decision Making and Path Planning Approach for Navigation in Complex Traffic, Ensuring Collision Avoidance and Seamless Driving Support.





BACKGROUND



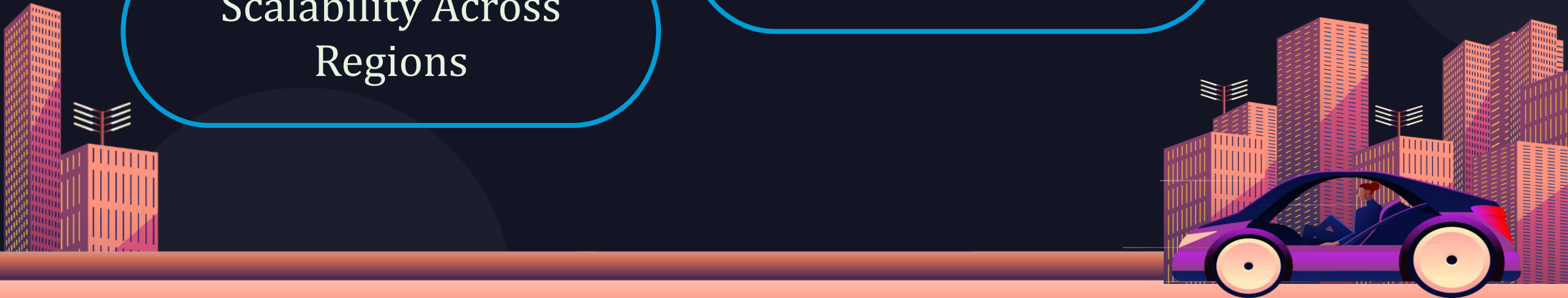
Why Advanced Decision Making is important?

Complex Traffic
Scenarios

Safety Enhancement

Human Like Decision
Making

Scalability Across
Regions



RESEARCH GAP

Research 01

Path Planning for Autonomous Vehicles using Model Predictive Control

Research 02

“Risk Assessment and Mitigation in Local Path Planning for Autonomous Vehicles With LSTM Based Predictive Model

Research 03

Multipolicy Decision Making in Dynamic, Uncertain Environments for Autonomous Driving

Research 04

Deep Reinforcement Learning for Human Like Driving Policies in Collision Avoidance Tasks of Self Driving Cars

Research 05

Path planning and integrated collision avoidance for autonomous vehicles

RESEARCH GAP

Features	Research 1	Research 2	Research 3	Research 4	Research 5	MANTHRA-X
Adaptive Decision Making	No	Yes	Yes	Yes	Yes	
Real Time Path Planning	Yes	Yes	No	No	Yes	
Collision Avoidance Mechanism	Yes	Yes	No	Yes	Yes	
Human Like Behavior	No	No	No	Yes	No	
Complex and Dynamic Traffic Interaction	No	No	Yes	No	No	

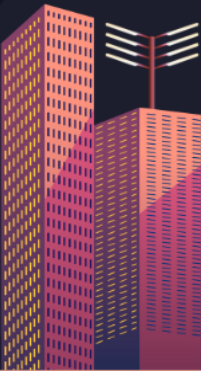
RESEARCH PROBLEM

What are the key challenges in ensuring reliable decision making for vehicles interacting with multiple agents?

Why is it important to accurately model and mimic human driving behaviors in autonomous systems?

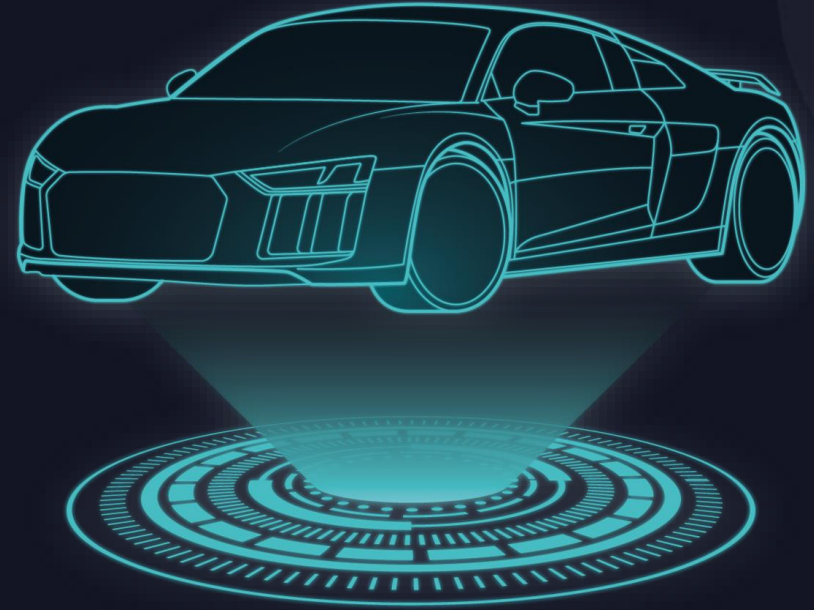
How can systems that adapt to unpredictable vehicle interactions and environmental conditions in real time?

How can decision making and path planning algorithms be tailored to meet the unique challenges of dynamic and unstructured traffic conditions



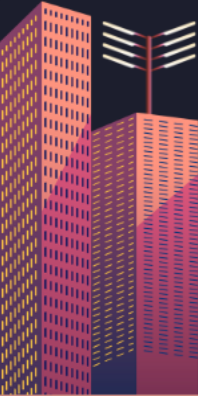
SPECIFIC OBJECTIVE

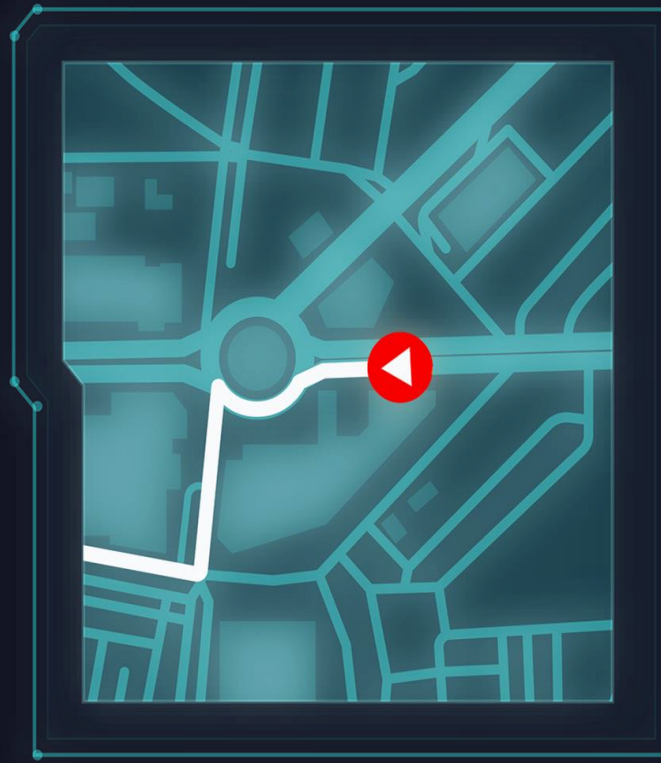
Develop and integrate advanced decision making and path planning capabilities for vehicles to improve collision avoidance and adaptability in dynamic traffic environments, with a particular focus on addressing the unique driving conditions.



SUB OBJECTIVES

- Dynamic human behavior adaptation
- Adaptive multi agent learning for mixed traffic environments
- Create a decision making system that can explore different future actions and outcomes in real time
- Predict potential collision risks based on the surrounding vehicles

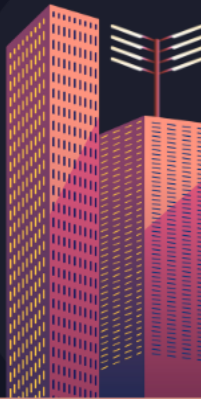
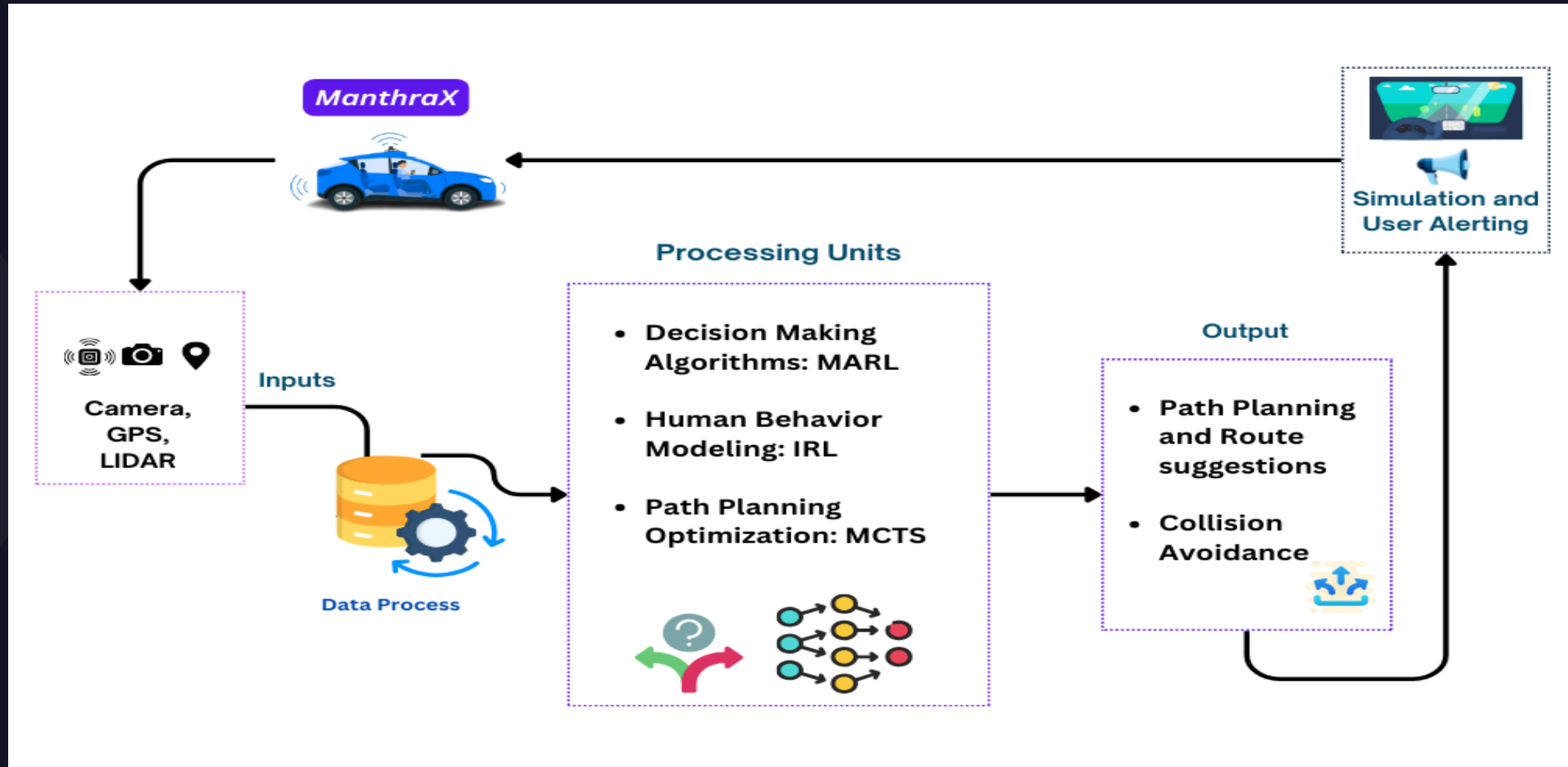




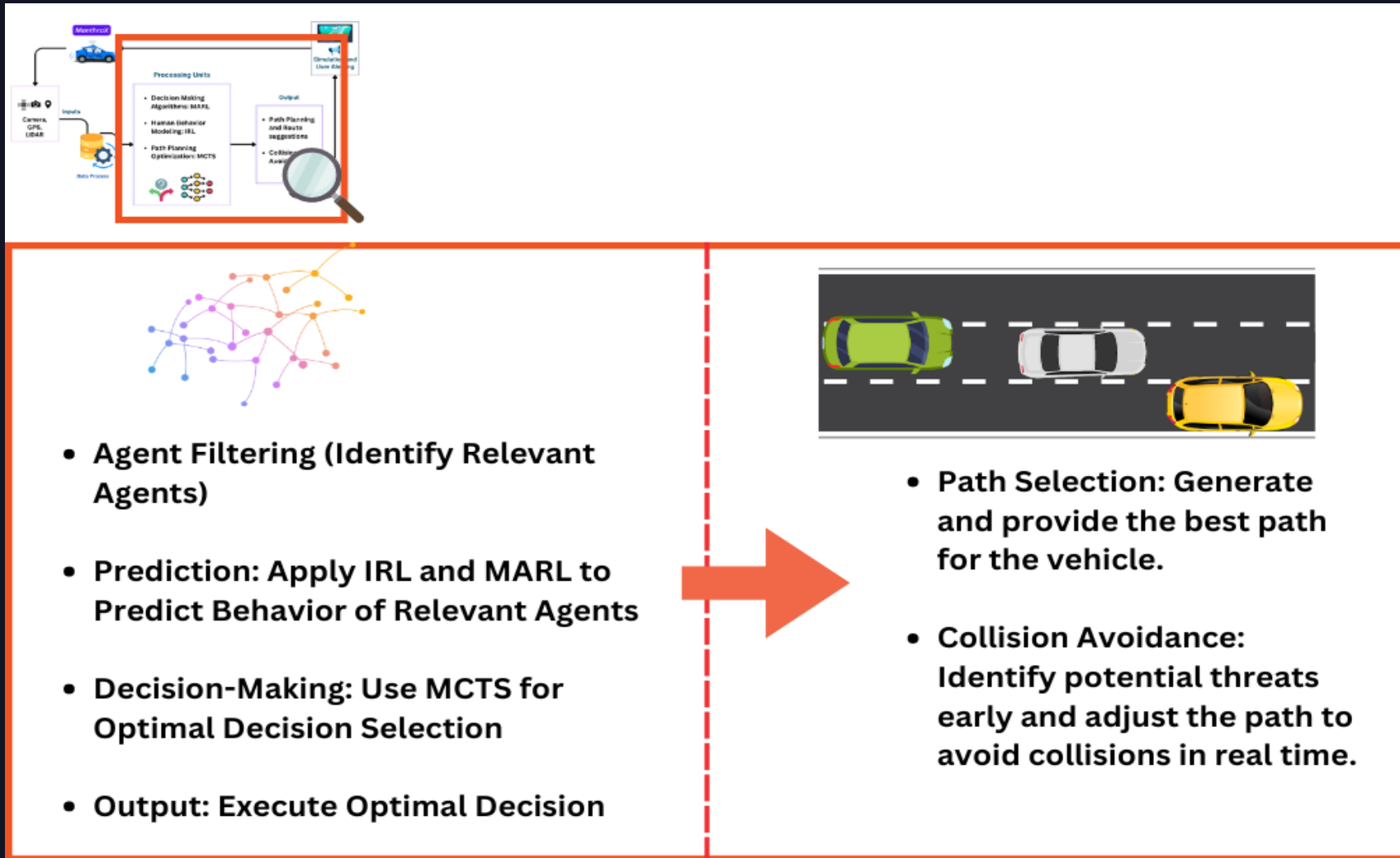
METHODOLOGY



SYSTEM DIAGRAM



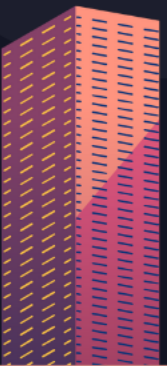
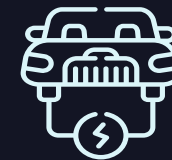
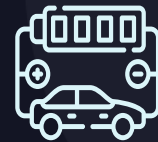
SYSTEM DIAGRAM



REQUIREMENT SPECIFICATION

System Requirements

- **High-Performance Computing (HPC) Resources** : HPC clusters or workstations for processing complex algorithms and run the simulation.
- **Sensors** : Cameras (for visual data), LIDAR (for depth and distance measurements), GPS (for location tracking)
- **Networking** : High speed internet connection for real time data transfer and communication with external systems.
- **Cloud Storage**: For storing large datasets, including historical decision-making scenarios and real-time logs.



REQUIREMENT SPECIFICATION

Software Requirements

Development Tools:

IDEs: VSCode, PyCharm, Jupyter Notebook
Version Control: Git(GitHub, GitLab)

Frameworks and Libraries:

AI/ML: Tensorflow, Pytorch, Keras, Scikit-learn, Pgmpy.

Cloud Services:

Microsoft Azure Services

Databases:

NoSQL Databases: MongoDB, Cassandra

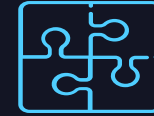
❑ Simulation Frameworks:

CARLA, AirSim, or Unity

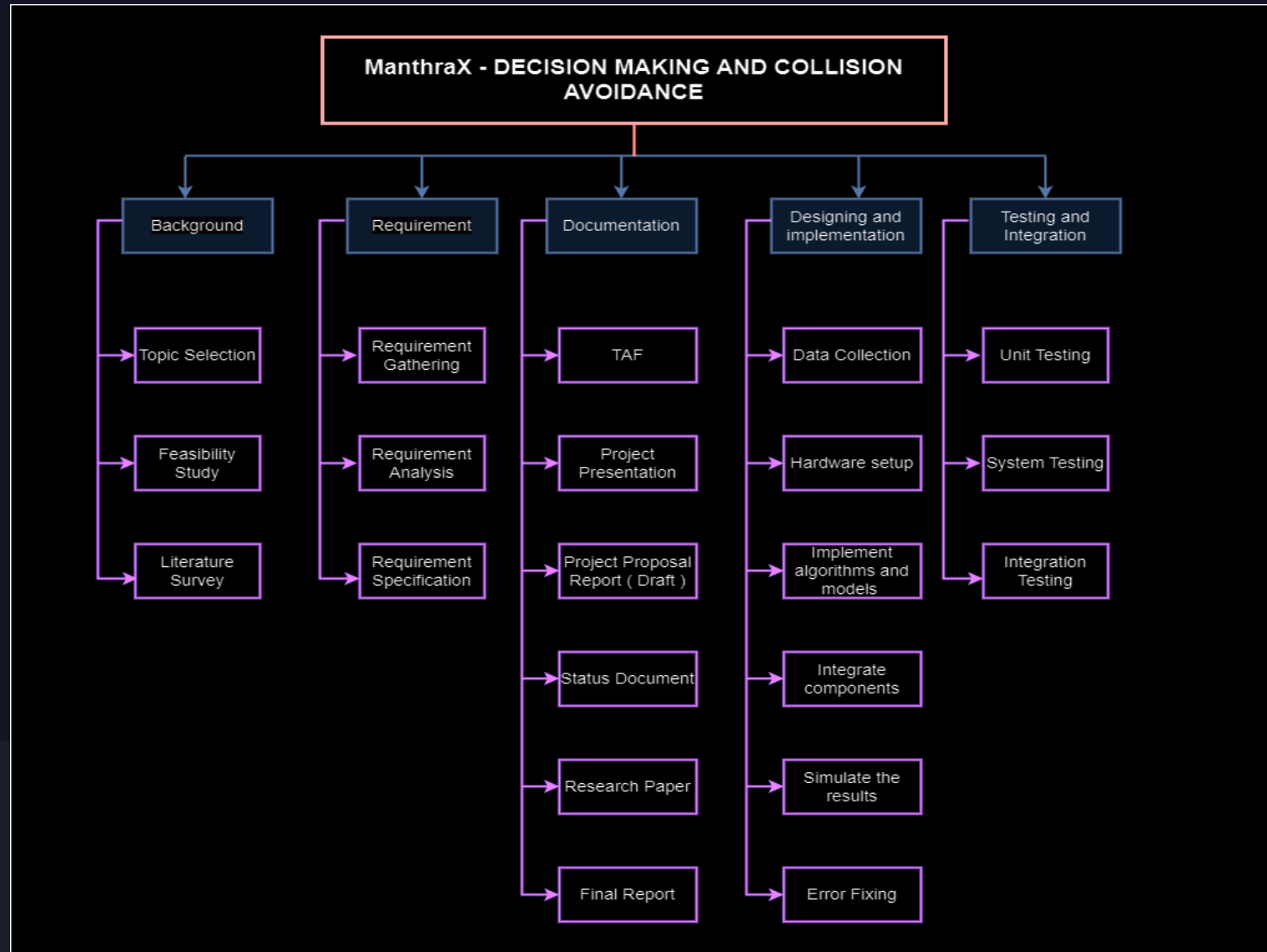
REQUIREMENT SPECIFICATION

Personal Requirements

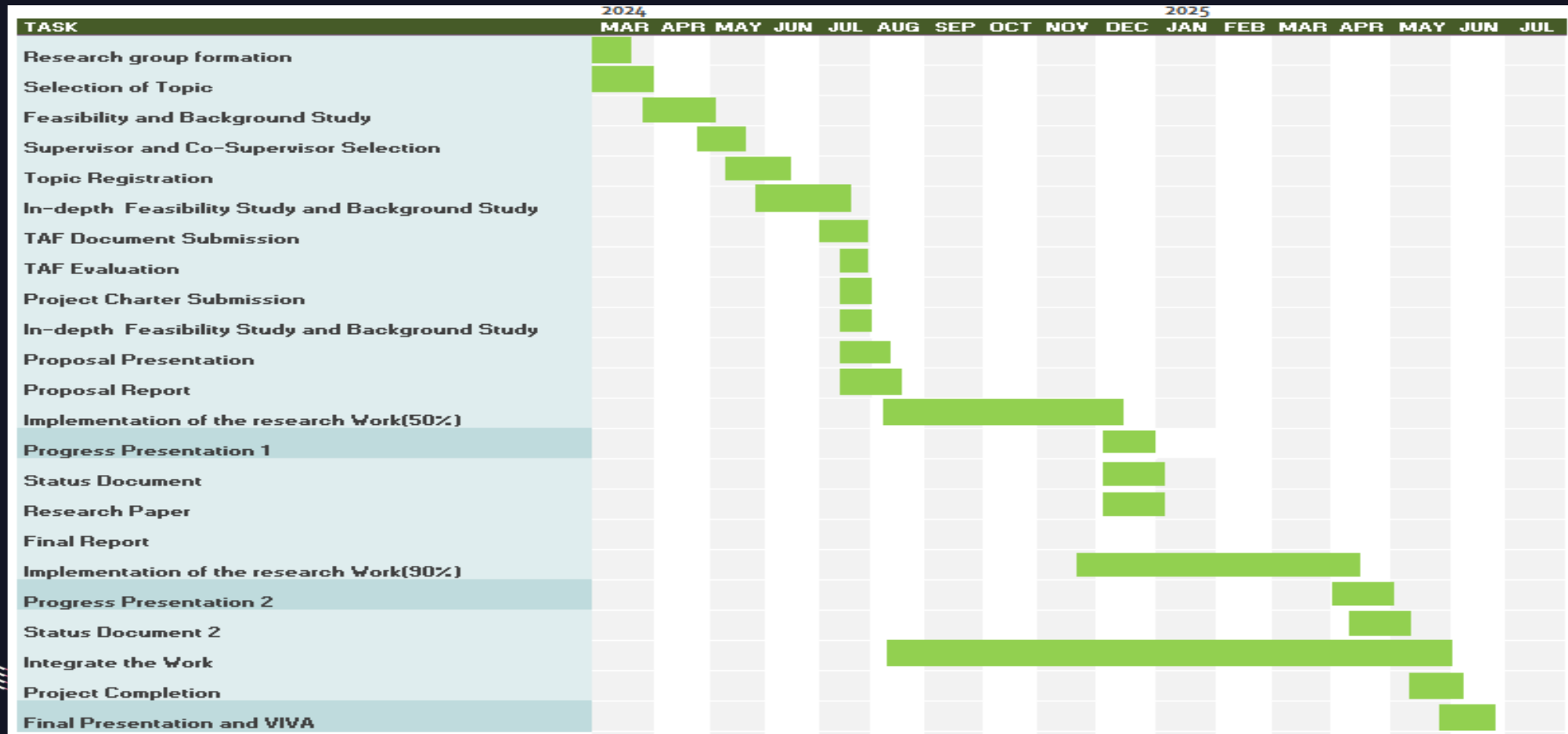
- Technical Proficiency
- Experience with Tools and Frameworks
- Problem Solving Abilities
- Teamwork and Communication
- Project Management
- Adaptability
- Time Management Skills



WORK BREAKDOWN CHART



GANNT CHART



References

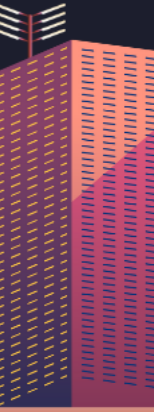
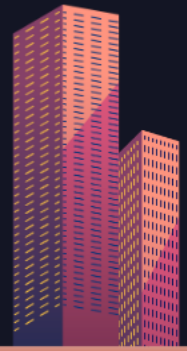
- [1] C. Liu, S. Lee, S. Varnhagen and H. E. Tseng, "Path planning for autonomous vehicles using model predictive control," 2017 IEEE Intelligent Vehicles Symposium (IV), Los Angeles, CA, USA, 2017, pp. 174-179, doi: 10.1109/IVS.2017.7995716.
- [2] H. Wang et al., "Risk Assessment and Mitigation in Local Path Planning for Autonomous Vehicles With LSTM Based Predictive Model," in IEEE Transactions on Automation Science and Engineering, vol. 19, no. 4, pp. 2738-2749, Oct. 2022, doi: 10.1109/TASE.2021.3075773.
- [3] A. G. Cunningham, E. Galceran, R. M. Eustice and E. Olson, "MPDM: Multipolicy decision-making in dynamic, uncertain environments for autonomous driving," 2015 IEEE International Conference on Robotics and Automation (ICRA), Seattle, WA, USA, 2015, pp. 1670-1677, doi: 10.1109/ICRA.2015.7139412.
- [4] R. Emuna, A. Borowsky, and A. Biess, "Deep reinforcement learning for human-like driving policies in collision avoidance tasks of self-driving cars," arXiv preprint arXiv:2006.04218, 2020.
- [5] K. Berntorp, "Path planning and integrated collision avoidance for autonomous vehicles," 2017 American Control Conference (ACC), Seattle, WA, USA, 2017, pp. 4023-4028, doi: 10.23919/ACC.2017.7963572.





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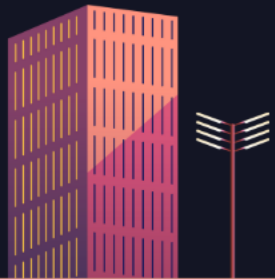


Component 03

Ethical Decision Making in Autonomous Vehicles



Enhance Ethical Decision Making by Integrating Ethical Frameworks, Driver Emotions and Object Prioritization.



BACKGROUND



RESEARCH GAP

Research 01

“Incorporating Ethical Considerations Into Automated Vehicle Control”

Research 02

“Decision-Making Technology for Autonomous Vehicles: Learning-Based Methods, Applications and Future Outlook”

Research 03

“Large Language Models for Autonomous Driving: Real-World Experiments”

Research 04

“Ethical Decision Making in Autonomous Vehicles: The AV Ethics Project”

Research 05

“Emotion detection and face recognition of drivers in autonomous vehicles in IoT platform”

RESEARCH GAP

Features	Research 1	Research 2	Research 3	Research 4	Research 5	MANTHRA-X
Personalized Ethical Decision-Making	No	No	Yes	No	No	✓
Integration of Real-Time Driver Emotion into Ethical Decision Making	No	No	Yes	No	Yes	✓
Culturally sensitive Ethical Decision making	No	No	No	No	No	✓
Sophisticated Object Prioritization	No	No	No	Limited	No	✓
Ethical Framework Consistency Across Scenarios	Yes	Yes	Yes	No	Yes	✓
Multi-Criteria Decision-Making Approaches	No	No	No	Yes	Yes	

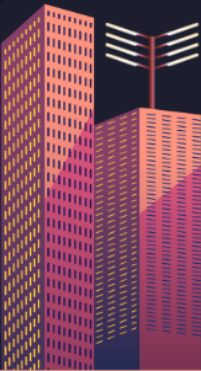
RESEARCH PROBLEM

How can ethical decision-making models be developed that accurately reflect individual driver preferences and emotional states?

What factors should be considered when prioritizing objects in complex driving scenarios, and how can these be integrated?

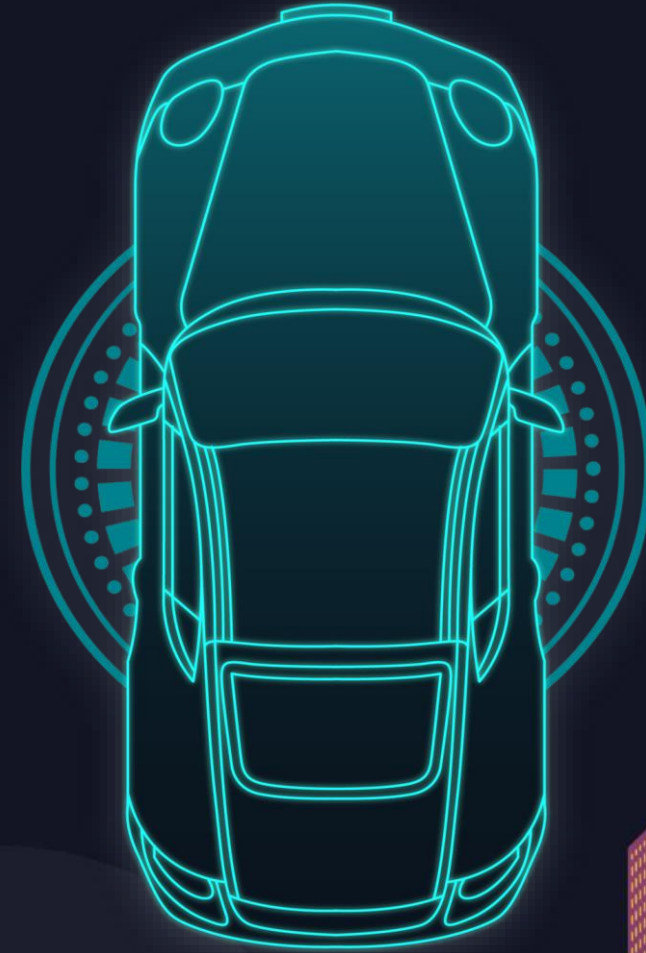
What facial features and expressions are most indicative of driver emotional states relevant to driving?

How can vocal cues be used to assist autonomous vehicles in making ethical decisions by interpreting driver intent, emotional state, and workload?



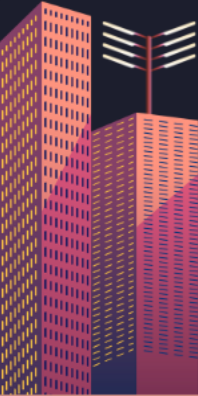
SPECIFIC OBJECTIVE

To develop a comprehensive ethical decision-making system for autonomous vehicles that prioritizes human safety, incorporates driver preferences, and adapts to dynamic environments, with a focus on ethical frameworks, object prioritization and personalized ethical considerations.



SUB OBJECTIVES

- Incorporate and Operationalize Ethical Principles and Frameworks
- Develop and Integrate Driver Emotion Recognition
- Establish Object and Road User Prioritization Techniques
- Ensure System Adaptability to Dynamic and Diverse Environments
- Enhance Transparency and User Trust



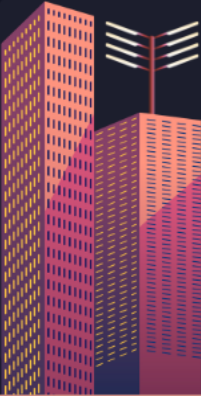
METHODOLOGY

SYSTEM DIAGRAM

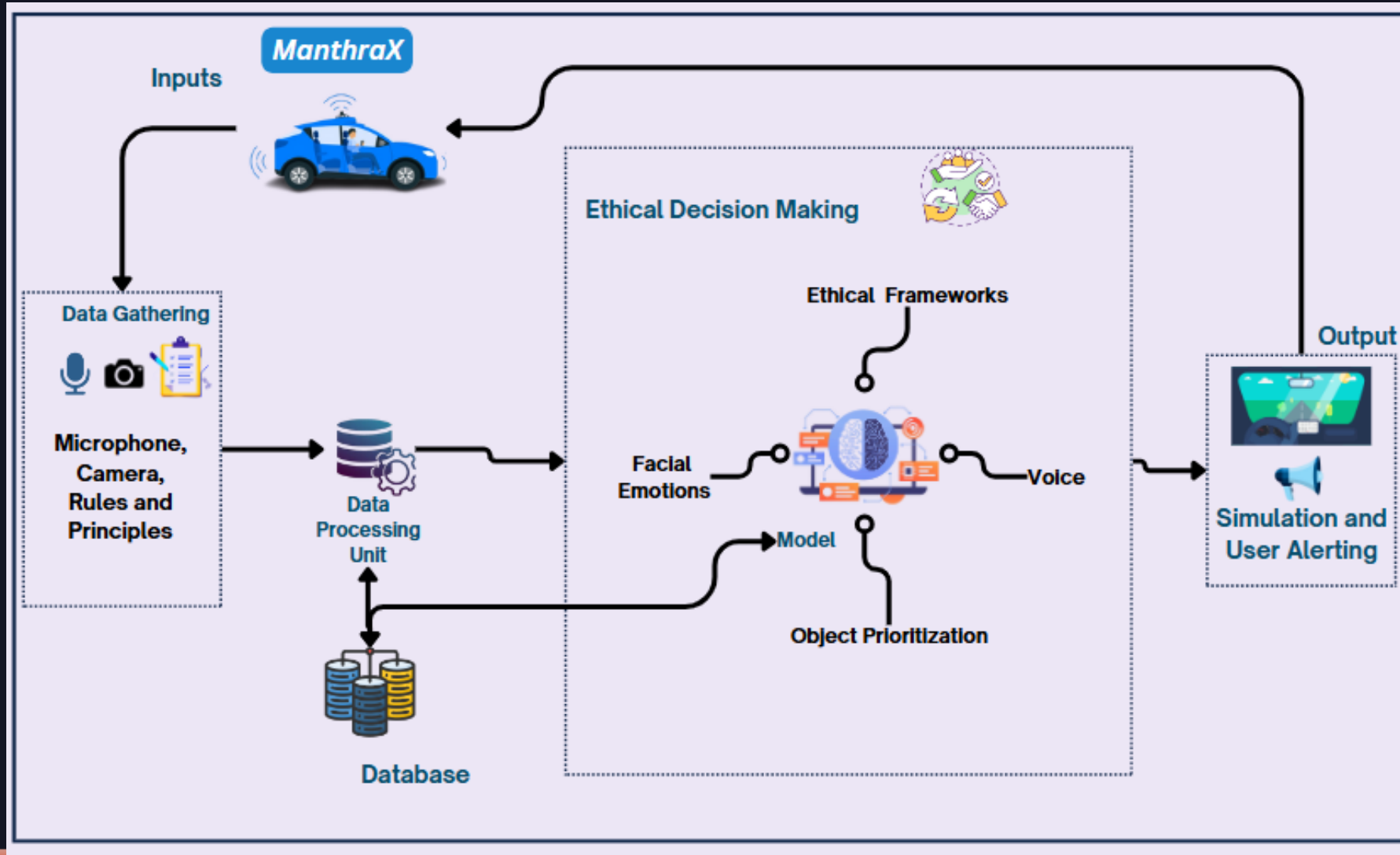
REQUIREMENT SPECIFICATION

- System Requirements
- Software Requirements
- Personal Requirements

TECHNOLOGIES TO BE USED



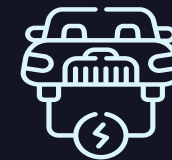
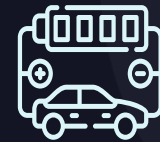
SYSTEM DIAGRAM



REQUIREMENT SPECIFICATION

System Requirements

- **High-performance Computing (HPC) Cluster or Workstations:** For processing complex algorithms related to ethical decision-making and real-time data analysis.
- **High-definition Cameras:** For capturing the driver's emotional state and external environment.
- **Microphones:** For capturing audio cues that may contribute to understanding the driver's state or external conditions.
- **High-speed Internet Connection:** For real-time data transfer, communication with external systems, and updates.
- **Cloud Storage:** For storing large datasets, including historical decision-making scenarios and real-time logs.



REQUIREMENT SPECIFICATION

Software Requirements

☐ Development Tools:

IDEs: VSCode, PyCharm, Jupyter Notebook
Version Control: Git(GitHub, GitLab)

☐ Frameworks and Libraries:

AI/ML: Tensorflow, Pytorch, Keras
Emotion Detection: OpenFace, Vokaturi
Other : OpenCV

☐ Cloud Services:

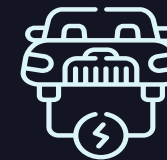
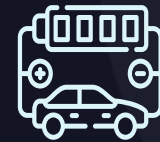
Microsoft Azure for Computing, storage, and machine Learning

☐ Databases:

NoSQL Databases: MongoDB, Cassandra

☐ Simulation Frameworks:

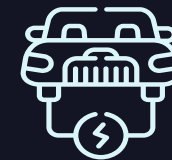
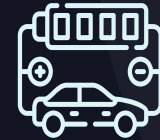
CARLA, AirSim, or Unity



REQUIREMENT SPECIFICATION

Personal Requirements

- Technical Proficiency
- Analytical Thinking
- Time Management Skills
- Commitment to Learning
- Teamwork and Communication
- Project Management
- Adaptability and Flexibility



TECHNOLOGIES TO BE USED



AI/ML Freameworks:
Tensorflow, Pytorch, Keras



Development Tools:
IDEs: VSCode, PyCharm,
Jupyter Notebook



Cloud Services:
Microsoft Azure



mongoDB

Databases:
NoSQL Databases:
MongoDB, Cassandra

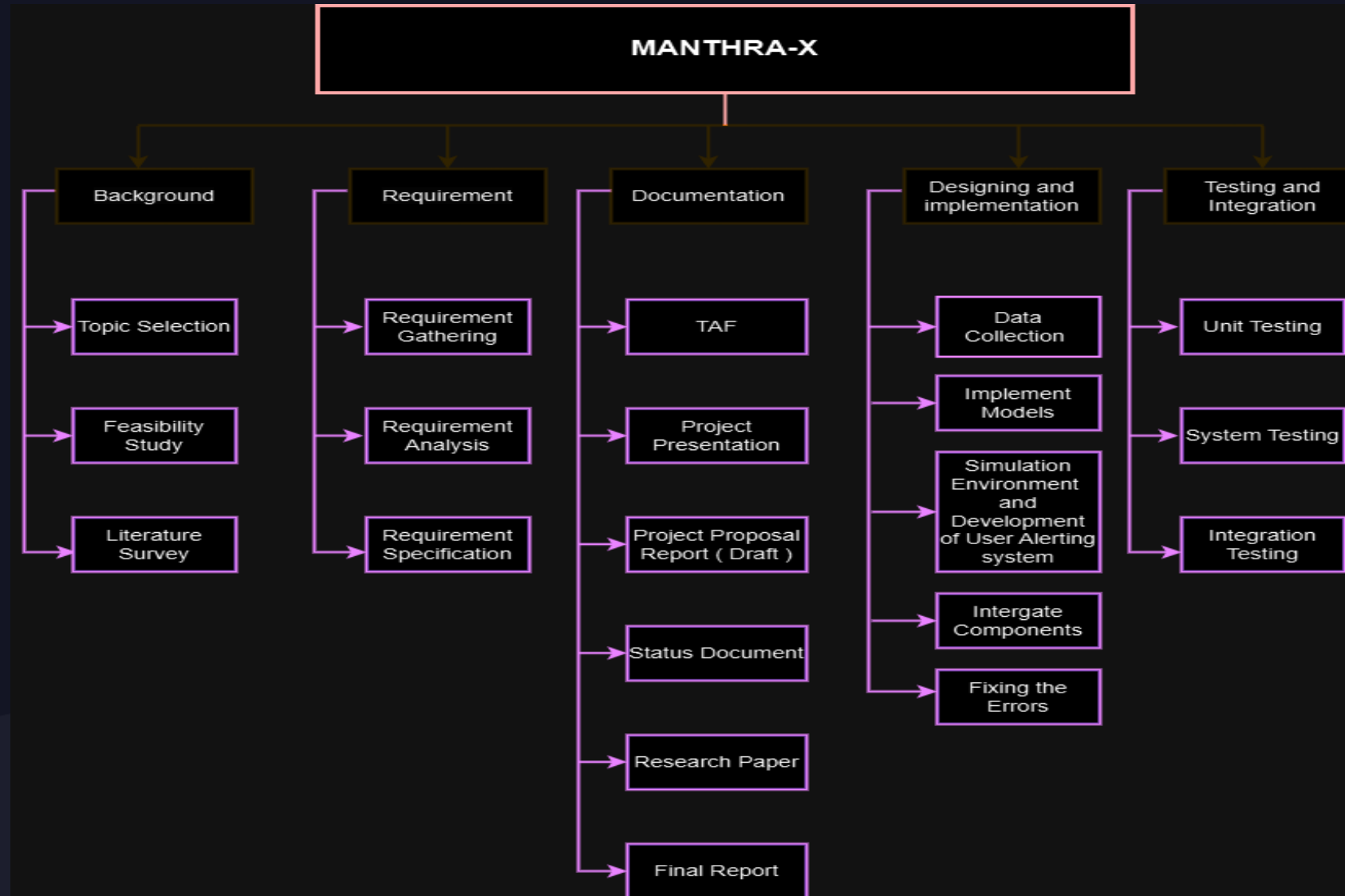


Simulation Frameworks:
CARLA, AirSim, or Unity

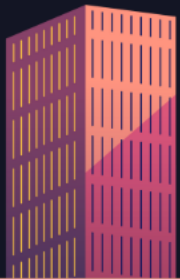
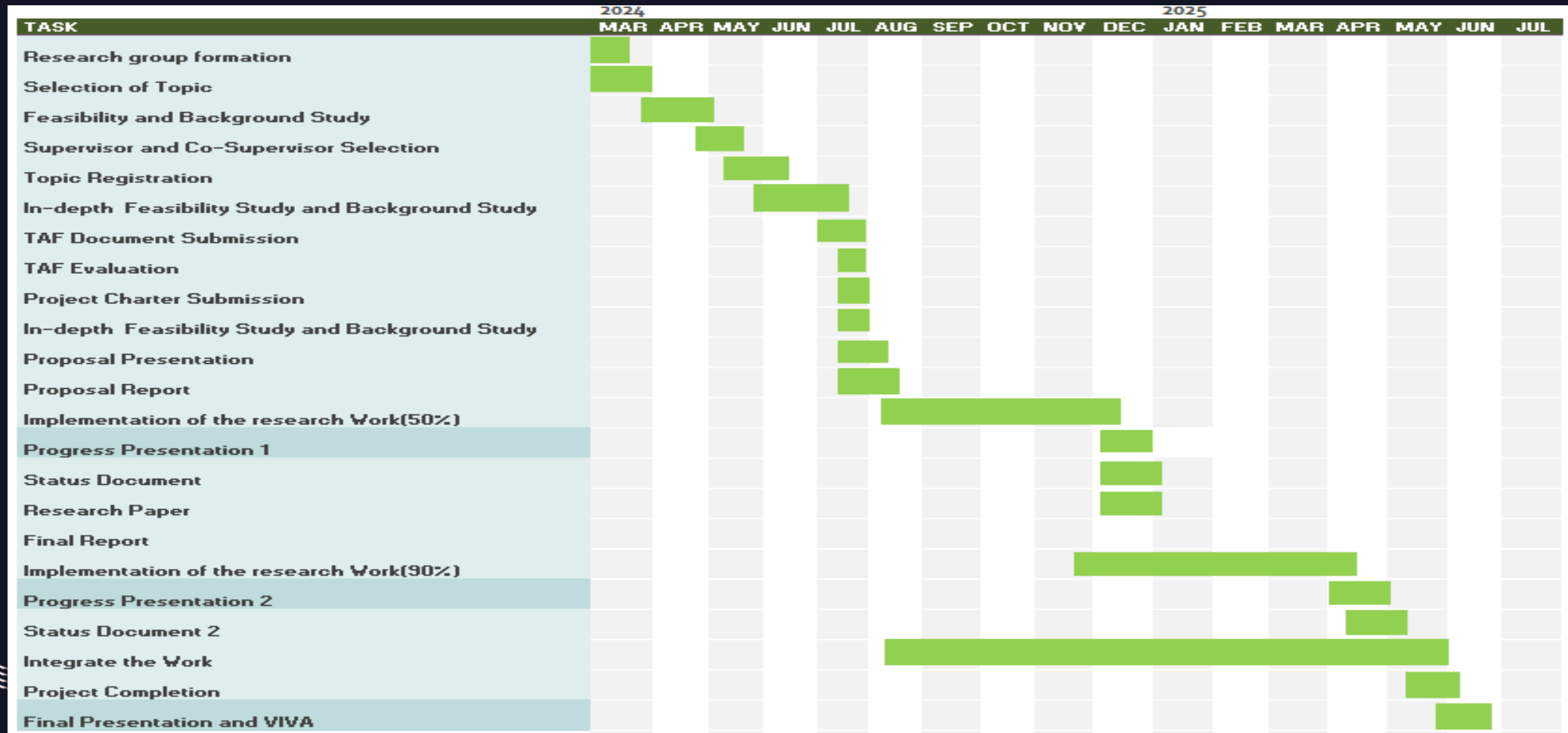


Web Frameworks: Node.js,
Flask, Django

WORK BREAKDOWN CHART



GANNT CHART



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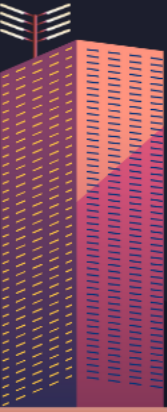
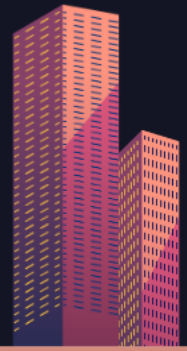
[Emotion detection and face recognition of drivers in autonomous vehicles in IoT platform - ScienceDirect](#)



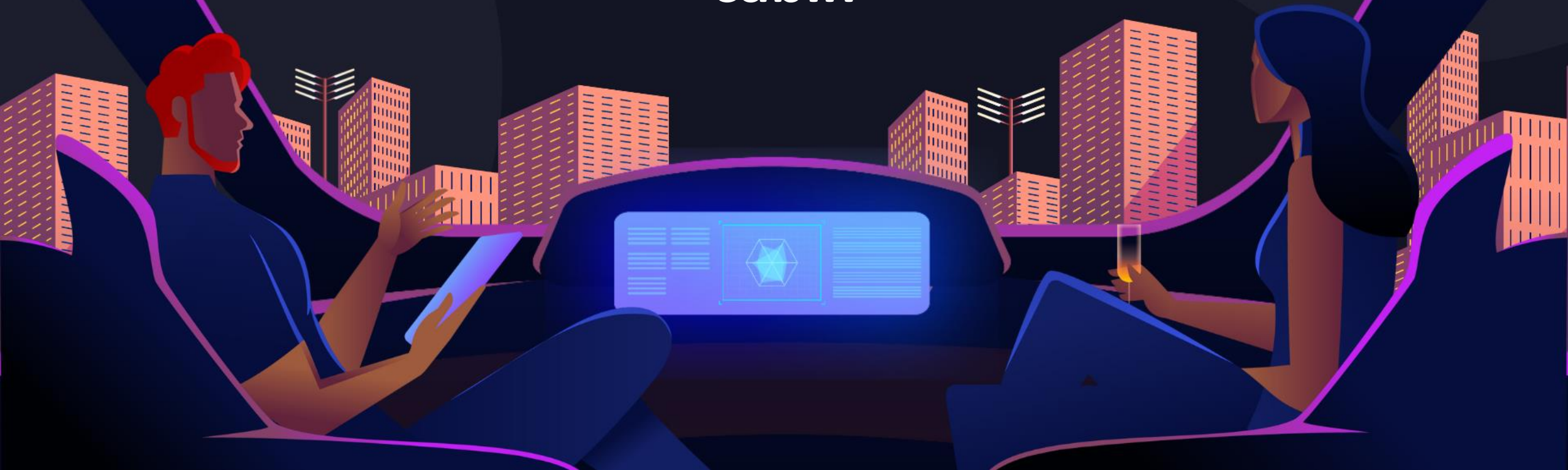


IT21174780 | D.M.M.I.T. DISSANAYAKA

BSc (Hons) Degree in Information Technology (specialization in Data Science)



Enhancing Autonomous Vehicle Safety in cabin



Background

Identifying and monitor security risks such as weapons and unauthorized items in the car cabin.

- Image Recognition: Use deep learning models to process real-time camera feeds and detect unauthorized items.
- Voice Analysis: Analyze audio inputs to identify suspicious or unauthorized behaviors through voice patterns.



Research Gap

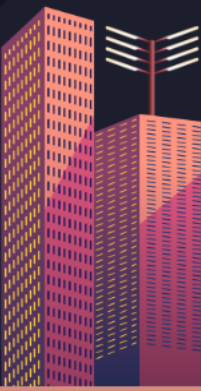
Enhancing Low-Light Resolution for In-Cabin Security in Autonomous Vehicles

Addressing Voice Overlap Challenges in Multi-Occupant Autonomous Vehicle Cabins

Enhancing the speed and accuracy of real-time data processing

False Positives and False Negatives

There is a need for systems that ensure privacy while still providing effective monitoring.



Competition comparison

Features	In-Cabin Monitoring for Avs[6]	sensor fusion in autonomous vehicles[4]	Autonomous Vehicles [5]	Security strategy for autonomous vehicle[7]	MANTHRA-X (proposed System)
Integration of Image and Voice Recognition					
False Positive/Negative Reduction in Security Systems					
Advanced Convolutional Neural Networks for Vehicle Security					
Voice Tone and sounds in Security Systems					
Ethical and Privacy Considerations					

Research Problem

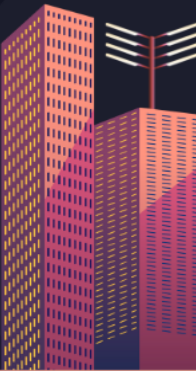
How to ensure accurate and reliable detection in autonomous vehicles to prevent security breaches ?

Synchronizing object and voice detection to accurately identify unauthorized activities.



How challenging is it to identify unauthorized items, such as weapons, in low-light scenarios?

Identifying unauthorized voices or keywords in an environment with multiple overlapping voices.



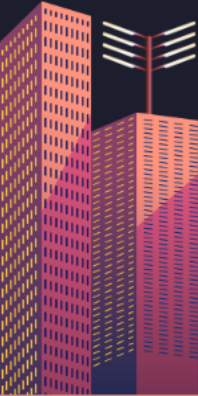
SPECIFIC OBJECTIVE

Implement an application to accurately identify and monitor security risks, such as weapons and unauthorized items, in the car cabin by integrating real-time camera feeds and audio inputs, ensuring reliable performance in every condition.

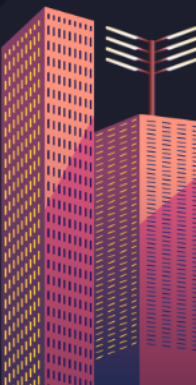
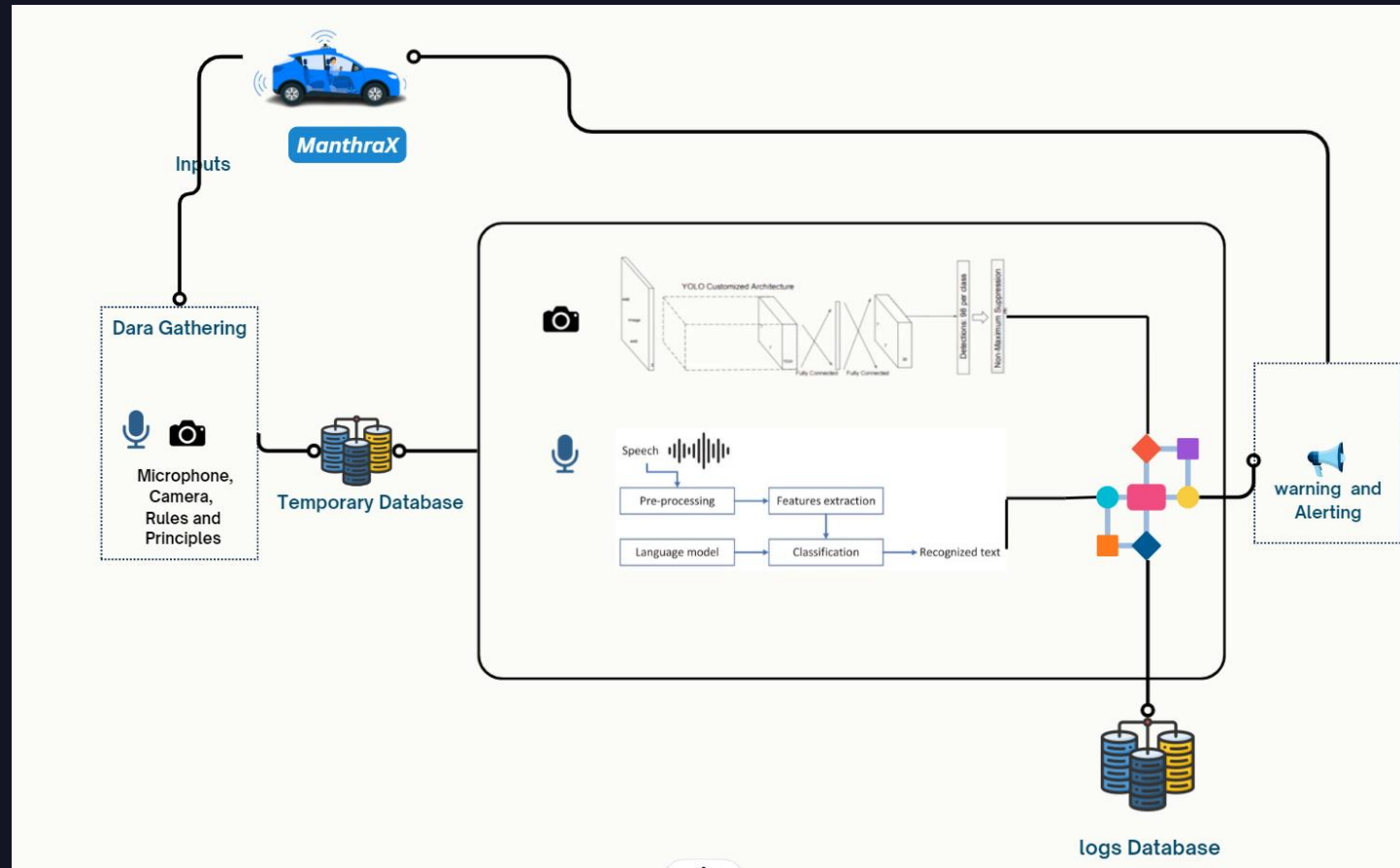


Sub Objectives

- Utilize advanced image and audio processing algorithms to analyze camera feeds and audio inputs continuously.
- Achieve high accuracy in detection while minimizing false positives.
- Protect the privacy and security of the data collected by the system.
- Design an intuitive and user-friendly interface for both drivers and passengers.



Methodology



Technologies to be Used



AI/ML Frameworks:
TensorFlow, PyTorch, Keras.



Cloud Services:
Microsoft Azure



Development Tools:
IDEs: VSCode, PyCharm,
Jupyter Notebook



Web Frameworks: Node.js,
Flask, Django



Security: Encryption.



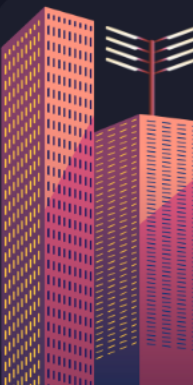
APIs: RESTful APIs



Databases:
NoSQL Databases:
MongoDB, Cassandra



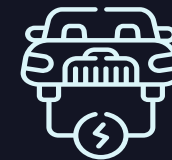
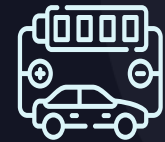
Simulation Frameworks:
CARLA, AirSim, or Unity



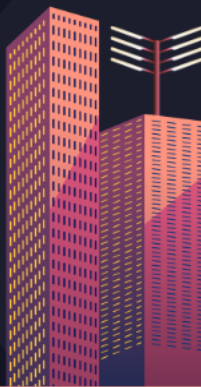
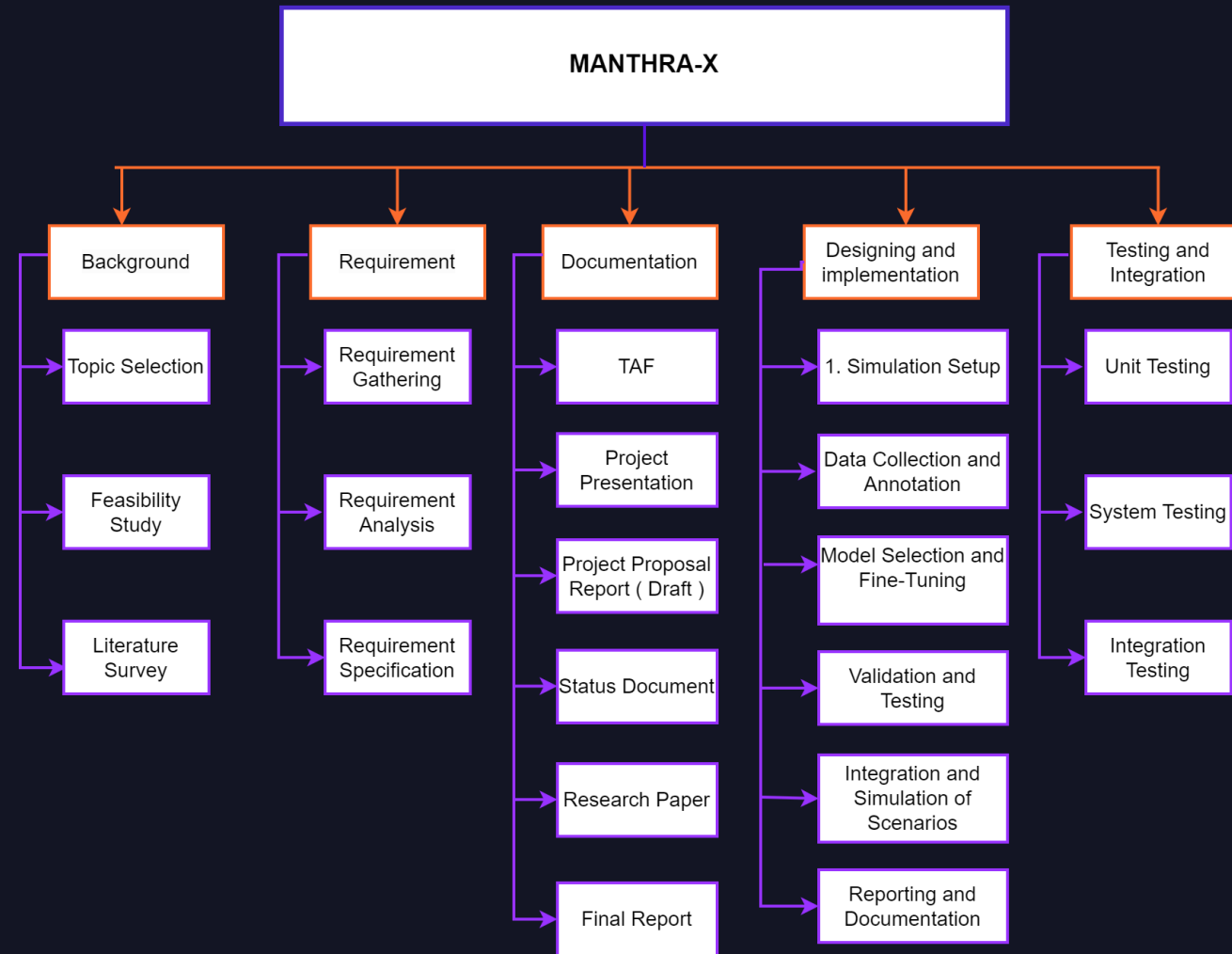
REQUIREMENT SPECIFICATION

System Requirements

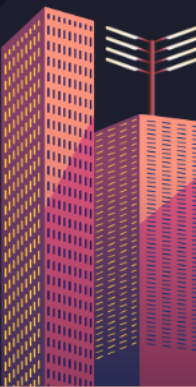
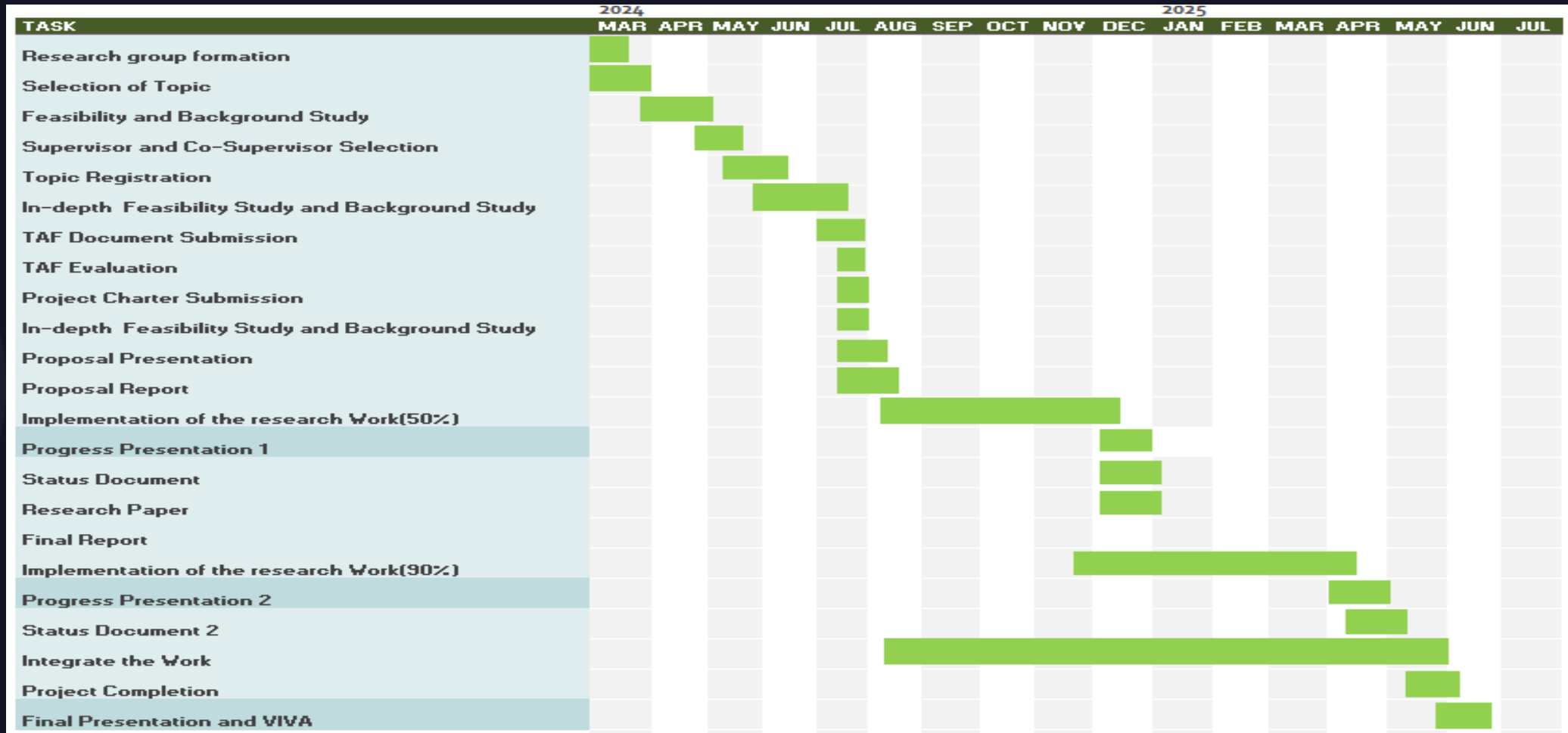
- ❖ **High-performance computing (HPC) cluster or workstations:** For model training, simulation, and real-time processing.
- ❖ **High-resolution cameras:** For capturing detailed visual information.
- ❖ **Microphones:** High-quality microphones are essential for capturing clear audio input.



Work Breakdown Structure (WBS)

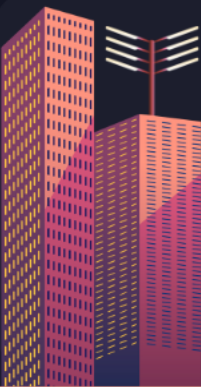


GANNT CHART



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- [4] Multimodal Sensor Fusion for Enhanced Security and Comfort in Autonomous Vehicles. (n.d.). Multimodal Sensor Fusion for Enhanced Security and Comfort in Autonomous Vehicles | IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9354376>
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- [6] A. Mishra, S. Lee, D. Kim, and S. Kim, "In-Cabin Monitoring System for Autonomous Vehicles," Sensors, vol. 22, no. 12, pp. 4360
- [7] A. A. Alsulami, Q. Abu Al-Haija, B. Alturki, A. Alqahtani, and R. Alsini, "Security strategy for autonomous vehicle cyber-physical systems using transfer learning," Journal of Cloud Computing: Advances, Systems and Applications, vol. 12, no. 1, pp. 1-18, 2023.



SUB OBJECTIVES

- **Incorporate and Operationalize Ethical Principles and Frameworks-**

Identify and integrate relevant ethical principles (e.g., utilitarianism, deontological ethics) into the system's decision-making processes to ensure alignment with established ethical standards.

- **Develop and Integrate Driver Emotion Recognition-**

Create and implement a system using voice and facial recognition technologies to accurately monitor and interpret the driver's emotional state, ensuring this information is effectively used in ethical decision-making.

- **Establish Object and Road User Prioritization Techniques-**

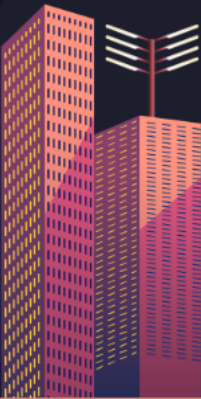
Develop methods for prioritizing objects and vulnerable road users based on ethical considerations and contextual information, ensuring decisions are aligned with safety and ethical standards.

- **Ensure System Adaptability to Dynamic and Diverse Environments –**

Design mechanisms that allow the system to adapt to varying driving conditions, societal values, and situational demands, maintaining ethical and safety standards.

- **Enhance Transparency and User Trust-**

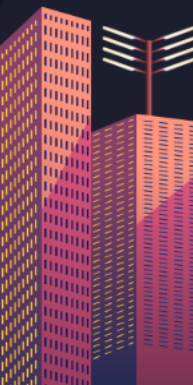
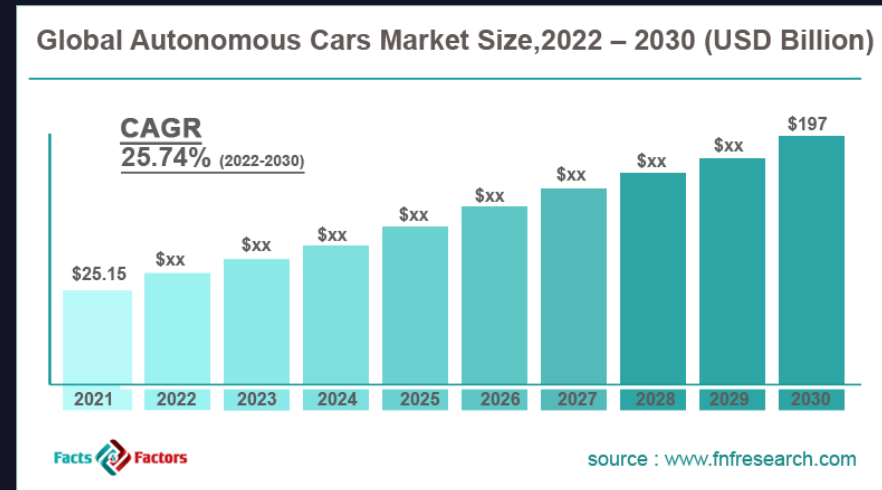
Create features that provide clear explanations of the system's decision-making processes, improving transparency and fostering trust among users regarding the ethical decisions made by the vehicle.



Market Analysis

1. Autonomous Vehicle Market Overview:

- The global autonomous vehicle market is projected to grow significantly, with a CAGR of 25.74% from 2022 to 2030. The demand is driven by technological advancements, increased safety concerns, and the potential for reducing traffic congestion.
- Key players in the market include Tesla, Waymo, Cruise, and Uber, with significant investments in R&D and collaborations to improve autonomous driving technologies.



Commercialization

Market Potential

```
graph LR; A[Market Potential] --- B[Demand Drivers: Increased safety and efficiency in traffic management.]; A --- C[Urban Mobility: Rising demand for advanced transportation solutions.]; A --- D[Emerging Economies: High growth potential in developing countries.];
```

Demand Drivers: Increased safety and efficiency in traffic management.

Urban Mobility: Rising demand for advanced transportation solutions.

Emerging Economies: High growth potential in developing countries.

Commercialization

Partnerships

- **Local Governments:** Collaborate on pilot programs and regulatory approvals.
- **Automotive Manufacturers:** Partner for technology integration and scaling production.
- **Tech Companies:** Work with AI and sensor technology firms for innovation and development.
- **Academic Institutions:** Engage with universities for research support and testing in real world scenarios.



Thank you

